

Entry diagnostic — the Grade 9 toolkit

One lesson to find out exactly what each learner still owns from Grade 9, before anything new is built on it.

LESSON 1

1 × 40 MIN

ALGEBRA 10, REVISION

P1 · 1.1–1.8

LESSON CLOCK

3'

30'

5'

2'

Explain the purpose — this is not for a mark	3 min
The paper	30 min
Self-mark against the answers	5 min
Record the two weakest sections	2 min

LEARNING OBJECTIVES

- Factorise a quadratic and solve it.
- Use the formula and the discriminant.
- Simplify surds and index expressions.
- Read a learner’s gaps from the diagnostic and record them.

TEXTBOOK

National	Повторение, pp. 6–23
Cambridge	Pure Mathematics 1, pp. 1–23
Workbook	Diagnostic paper

01

Explanation

Why the year starts here

Grade 10 algebra spends its whole first quarter on **functions**. Almost every example is a quadratic, a linear expression or a surd. A learner who cannot factorise $x^2 - 5x + 6$ quickly will not be thinking about domain and range — they will still be thinking about factorising.

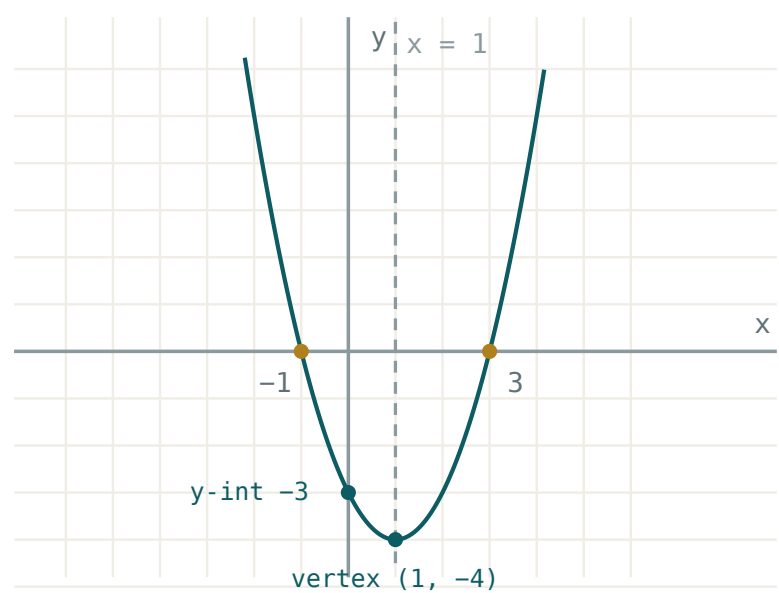
WHAT THIS LESSON IS FOR

Not a mark. A map. By the end of the lesson every learner should be able to name the **two** sections they were weakest in, and those two become their private revision list for the quarter.

Mark it together in the last five minutes, out of 20, in six sections of roughly equal weight.

The six sections

§	SKILL	MARKS	WHERE IT IS NEEDED THIS QUARTER
A	factorising a quadratic	4	roots, domain of a fraction
B	the quadratic formula and D	4	range, number of roots
C	surds and indices	3	domain of a root function
D	straight-line graphs	3	linear modelling
E	sketching a parabola	3	quadratic modelling
F	inequalities on a number line	3	writing a range



Everything in section E: the roots, the axis of symmetry, the vertex and the y-intercept of one parabola.

SAY THIS OUT LOUD

A low score today predicts nothing about the year. It only says which twenty minutes of revision will pay for themselves fastest.

02 Worked examples

EXAMPLE 1 Solve $x^2 - 5x + 6 = 0$ by factorising.

- ① Two numbers multiplying to 6 and adding to -5 .
They are -2 and -3 .

② $(x - 2)(x - 3) = 0$

③ $x = 2$ or $x = 3$

ANSWER

$x = 2, x = 3$

EXAMPLE 2 Find D for $2x^2 - 4x + 5 = 0$ and say how many real roots there are.

- ① $a = 2, b = -4, c = 5$
Read the signs carefully.

② $D = 16 - 4 \cdot 2 \cdot 5 = 16 - 40$

- ③ $D = -24 < 0$
No real roots.

ANSWER

$D = -24$, no real roots

EXAMPLE 3 Simplify $\sqrt{50} + \sqrt{18}$.

① $\sqrt{50} = 5\sqrt{2}$
 $50 = 25 \cdot 2.$

② $\sqrt{18} = 3\sqrt{2}$
 $18 = 9 \cdot 2.$

③ $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$
Like terms.

ANSWER

$$8\sqrt{2}$$

03 Interactive model

Work one question from each section on the board while the class self-marks.

Six questions, one from each section

$x^2 + 2x - 15$ factorises to:

For $x^2 - 6x + 9 = 0$, D is:

$\sqrt{27}$ in simplest form is:

The gradient of $2x + y = 6$ is:

2

-2

6

$\frac{1}{2}$

The vertex of $y = (x - 3)^2 + 1$ is:

$(3, 1)$

$(-3, 1)$

$(3, -1)$

$(1, 3)$

$-2x > 8$ gives:

$x > -4$

$x < -4$

$x > 4$

$x < 4$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Diagnostic test	Diagnostik test	Диагностический тест
Prior knowledge	Oldingi bilim	Ранее изученное
Factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Quadratic equation	Kvadrat tenglama	Квадратное уравнение
Discriminant	Diskriminant	Дискриминант
Surd	Irratsional ifoda	Иррациональное выражение
Index (exponent)	Daraja ko'rsatkichi	Показатель степени
Gap	Bo'shliq	Пробел
Baseline	Boshlang'ich daraja	Исходный уровень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$x^2 - 9$ factorises to:

If $D > 0$ a quadratic has:

$\sqrt{8} \cdot \sqrt{2}$ equals:

The line through $(0, 3)$ with gradient 2 is:

$$y = 2x + 3$$

$$y = 3x + 2$$

$$y = 2x - 3$$

$$y = 3x$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x^2 - 7x + 12 = 0$

ANSWER $x = 3, x = 4$

2. Solve $x^2 - 16 = 0$

ANSWER $x = \pm 4$

3. Find D for $x^2 + 4x + 4 = 0$

ANSWER 0

4. Simplify $\sqrt{12}$

ANSWER $2\sqrt{3}$

5. Find the gradient of $y = 5 - 3x$

ANSWER -3

6. Where does $y = x^2 - 4$ cut the x -axis?

ANSWER $(\pm 2, 0)$

7. Solve $3x + 2 < 11$

ANSWER $x < 3$

Medium · the standard question

7 PROBLEMS

1. Solve $2x^2 - 5x - 3 = 0$

ANSWER $x = 3, x = -0.5$

2. Solve $x^2 + x - 12 = 0$

ANSWER $x = 3, x = -4$

3. Find D for $3x^2 - 2x + 1 = 0$ and say how many roots.

ANSWER $D = -8$, none

4. Simplify $\sqrt{75} - \sqrt{27}$

ANSWER $2\sqrt{3}$

5. Find the equation of the line through $(1, 5)$ and $(3, 11)$.

ANSWER $y = 3x + 2$

6. Write $y = x^2 - 6x + 5$ in completed-square form.

ANSWER $(x - 3)^2 - 4$

7. Solve $4 - 2x \geq 10$

ANSWER $x \leq -3$

Hard · stretch and reason

7 PROBLEMS

1. Solve $6x^2 + x - 2 = 0$

ANSWER $x = \frac{1}{2}, x = -\frac{2}{3}$

2. For what k does $x^2 + kx + 9 = 0$ have exactly one root?

ANSWER $k = \pm 6$

3. Simplify $\frac{6}{\sqrt{3}}$

ANSWER $2\sqrt{3}$

4. Find the vertex and roots of $y = x^2 - 4x + 3$.

ANSWER vertex $(2, -1)$, roots 1 and 3

5. Solve $x^2 - 5x + 6 > 0$

ANSWER $x < 2$ or $x > 3$

6. The line $y = 2x + k$ is a tangent to $y = x^2$. Find k .

ANSWER $x^2 - 2x - k = 0, D = 4 + 4k = 0$, so $k = -1$

7. Two numbers add to 12 and multiply to 35. Find them without trial.

ANSWER Roots of $t^2 - 12t + 35 = 0$: 5 and 7

07 Homework

Homework — 5 tasks

Only the two sections you were weakest in. Nothing else.

- Write down the two sections you scored lowest in.
- Five problems from the first of those two sections.
- Five problems from the second.
- Solve $x^2 - 3x - 10 = 0$ and check both roots by substitution.

5. Simplify $\sqrt{98} + \sqrt{50}$.

Quadratic equations

— factorising and the formula

Cambridge insert: the two methods, and knowing within five seconds which one this equation wants.

LESSON 2–3

2 × 40 MIN

ALGEBRA 10, REVISION

P1 · 1.1, 1.4–1.5

LESSON CLOCK

6'

14'

12'

6'

4'

Five equations — which method?	6 min
Factorising, including $a \neq 1$	14 min
The formula	12 min
Disguised quadratics	6 min
Homework	4 min

LEARNING OBJECTIVES

- Factorise a quadratic with leading coefficient 1 and with $a \neq 1$.
- Use the quadratic formula accurately.
- Choose between the two methods quickly.
- Solve equations that become quadratic after a substitution.

TEXTBOOK

National	Повторение, pp. 6–13
Cambridge	Pure Mathematics 1, pp. 2–14
Workbook	P1 Exercise 1A, 1D

01

Explanation

Two methods, one decision

THE ZERO PRODUCT RULE

If $A \cdot B = 0$ then $A = 0$ or $B = 0$. That is the whole reason factorising solves equations — and the reason the equation must be written with **zero on one side first**.

THE EQUATION LOOKS LIKE	USE
$x^2 + bx + c$ with small whole b, c	factorising — try it for five seconds
$ax^2 + bx + c$ with $a \neq 1$	factorising by splitting the middle term
ugly numbers, or the answer is asked for in surd form	the formula
no constant term	take out x — never divide by it

Factorising $2x^2 + 7x + 3$: find two numbers multiplying to $2 \cdot 3 = 6$ and adding to 7 — they are 1 and 6. Split the middle term and group:

$$2x^2 + x + 6x + 3 = x(2x + 1) + 3(2x + 1) = (2x + 1)(x + 3)$$

The formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

It never fails, and it is slower. Use it when factorising has not worked in five seconds, or when the question asks for an exact answer.

THREE PLACES MARKS ARE LOST

1. The whole of $-b$ is over the $2a$, not just the root.
2. $-b$ where b is already negative gives a **plus**.
3. $2a$, not 2, when $a \neq 1$.

Equations that become quadratic

Many equations are quadratics wearing a disguise. Substitute a letter for the repeated block and the disguise falls off:

$$x^4 - 5x^2 + 4 = 0 \text{ let } t = x^2 \implies t^2 - 5t + 4 = 0$$

Solve for t , then go back: $t = 1$ gives $x = \pm 1$, $t = 4$ gives $x = \pm 2$. Four roots.

The same trick handles 2^x , $\sqrt{x}$ and $\frac{1}{x}$ as the repeated block — and each time,

check the substitution allows the value before converting back.

02 Worked examples**EXAMPLE 1** Solve $3x^2 - 10x + 8 = 0$ by factorising.

- ① $3 \cdot 8 = 24$, and $-4 + (-6) = -10$
Two numbers multiplying to 24, adding to -10 .

- ② $3x^2 - 4x - 6x + 8 = 0$
Split the middle term.

- ③ $x(3x - 4) - 2(3x - 4) = 0$
Group.

- ④ $(3x - 4)(x - 2) = 0$

- ⑤ $x = \frac{4}{3}$ or $x = 2$

ANSWER

$$x = \frac{4}{3}, x = 2$$

EXAMPLE 2 Solve $2x^2 - 6x + 1 = 0$, giving exact answers.

$$\textcircled{1} \quad a = 2, b = -6, c = 1$$

$$\textcircled{2} \quad D = 36 - 8 = 28$$

$$\textcircled{3} \quad x = \frac{6 \pm \sqrt{28}}{4}$$

Note $-b = +6$.

$$\textcircled{4} \quad \sqrt{28} = 2\sqrt{7}$$

$$\textcircled{5} \quad x = \frac{6 \pm 2\sqrt{7}}{4} = \frac{3 \pm \sqrt{7}}{2}$$

Cancel the 2.

ANSWER

$$x = \frac{3 \pm \sqrt{7}}{2}$$

EXAMPLE 3 Solve $x^4 - 13x^2 + 36 = 0$.

① $t = x^2$
and $t \geq 0$.

② $t^2 - 13t + 36 = 0$

③ $(t - 4)(t - 9) = 0$

④ $t = 4$ or $t = 9$
Both are ≥ 0 ✓

⑤ $x = \pm 2$ or $x = \pm 3$

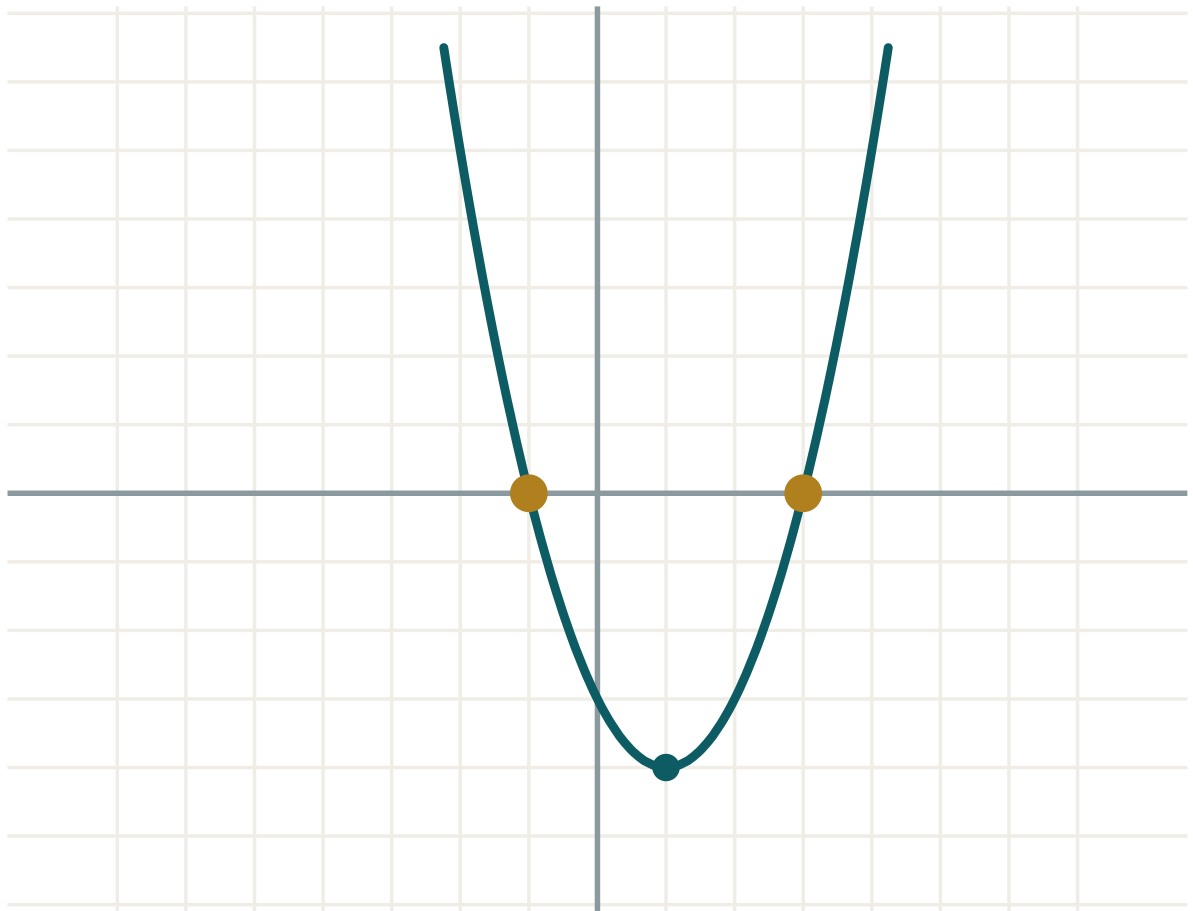
ANSWER

$$x = \pm 2, \pm 3$$

03 Interactive model

Slide the coefficients and watch the roots appear and disappear as the parabola lifts off the axis.

The roots of a quadratic



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Quadratic equation	Kvadrat tenglama	Квадратное уравнение
Coefficient	Koeffitsient	Коэффициент
Factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Root	Ildiz	Корень
Quadratic formula	Kvadrat tenglama formulasi	Формула корней
Zero product rule	Nolga ko'paytma qoidasi	Правило нулевого произведения
Substitution	O'rniga qo'yish	Подстановка
Disguised quadratic	Yashirin kvadrat tenglama	Уравнение, сводящееся к квадратному
Exact value	Aniq qiymat	Точное значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$x^2 - x - 6 = 0$ has roots:

2, -3

3, -2

6, -1

1, -6

For $2x^2 + 3x - 2 = 0$ the formula gives $2a$ equal to:

2

4

3

-2

$x^2 = 7x$ has roots:

7

0, 7

$\pm\sqrt{7}$

0

In $x^4 - 5x^2 + 4 = 0$, substituting $t = x^2$ gives how many values of x ?

Which method suits $x^2 - 4x + 1 = 0$?

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x^2 - 5x + 4 = 0$

ANSWER $x = 1, x = 4$

2. Solve $x^2 + 7x + 10 = 0$

ANSWER $x = -2, x = -5$

3. Solve $x^2 - 25 = 0$

ANSWER $x = \pm 5$

4. Solve $x^2 + 3x = 0$

ANSWER $x = 0, x = -3$

5. Solve $2x^2 - 8 = 0$

ANSWER $x = \pm 2$

6. Solve $x^2 - 2x + 1 = 0$

ANSWER $x = 1$ (repeated)

7. Solve $x^2 + x - 20 = 0$

ANSWER $x = 4, x = -5$

Medium · the standard question

7 PROBLEMS

1. Solve $2x^2 + 5x + 2 = 0$

ANSWER $x = -\frac{1}{2}, x = -2$

2. Solve $3x^2 - 7x + 2 = 0$

ANSWER $x = 2, x = \frac{1}{3}$

3. Solve $x^2 - 4x + 2 = 0$ exactly.

ANSWER $x = 2 \pm \sqrt{2}$

4. Solve $5x^2 - 3x = 0$

ANSWER $x = 0, x = 0.6$

5. Solve $x^4 - 10x^2 + 9 = 0$

ANSWER $x = \pm 1, \pm 3$

6. Solve $4x^2 - 12x + 9 = 0$

ANSWER $x = 1.5$ (repeated)

7. Solve $6x^2 - 5x - 6 = 0$

ANSWER $x = \frac{3}{2}, x = -\frac{2}{3}$

Hard · stretch and reason

7 PROBLEMS

1. Solve $3x^2 + 2x - 4 = 0$ exactly.

ANSWER $x = \frac{-1 \pm \sqrt{13}}{3}$

2. Solve $x^4 - 3x^2 - 4 = 0$

ANSWER $t = 4$ only (reject $t = -1$), so $x = \pm 2$

3. Solve $2^{2x} - 6 \cdot 2^x + 8 = 0$

ANSWER $2^x = 2$ or 4 , so $x = 1, 2$

4. Solve $x - 5\sqrt{x} + 6 = 0$

ANSWER $\sqrt{x} = 2$ or 3 , so $x = 4, 9$

5. The roots of $x^2 + px + 12 = 0$ differ by 1. Find p .

ANSWER $p = \pm 7$

6. Solve $\frac{1}{x^2} - \frac{5}{x} + 6 = 0$

ANSWER $t = \frac{1}{x}: t = 2, 3$, so $x = \frac{1}{2}, \frac{1}{3}$

7. Show that $x^2 + x + 1 = 0$ has no real root, without the formula.

ANSWER $(x + \frac{1}{2})^2 + \frac{3}{4}$ is a square plus a positive number, so never zero.

07 Homework

Homework — 6 problems

P1 Exercise 1A and 1D. Say which method you chose and why.

1. Solve $x^2 - 8x + 15 = 0$
2. Solve $2x^2 - 7x + 3 = 0$
3. Solve $x^2 - 6x + 4 = 0$ exactly.

4. Solve $3x^2 + 5x = 0$

5. Solve $x^4 - 5x^2 + 4 = 0$

6. Solve $x - 7\sqrt{x} + 12 = 0$

Completing the square and the discriminant

Cambridge insert: the form that hands you the vertex, and the number that counts the roots before you find them.

LESSON 4–5

2 × 40 MIN

ALGEBRA 10, REVISION

P1 · 1.3, 1.6

LESSON CLOCK

5'

14'

12'

5'

4'

Expand three squares to see the pattern backwards 5 min

Completing the square 14 min

What the form tells you 12 min

The discriminant 5 min

Homework 4 min

LEARNING OBJECTIVES

- Write $x^2 + bx + c$ in the form $(x + p)^2 + q$.
- Read the vertex and the least value off that form.
- Use D to say how many real roots there are.
- Find the values of a parameter that make a quadratic touch the axis.

TEXTBOOK

National Повторение, pp. 6–13

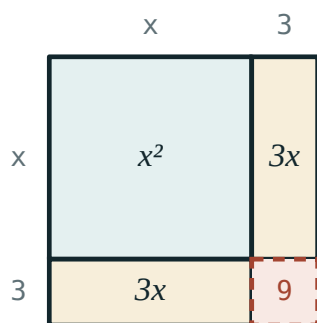
Cambridge Pure Mathematics 1, pp. 7–16

Workbook P1 Exercise 1B, 1F

01 Explanation

Completing the square

Expanding $(x + 3)^2 = x^2 + 6x + 9$ shows the pattern: the number inside is **half** the coefficient of x , and the constant is its square. Run it backwards and you can turn any $x^2 + bx$ into a square, at the cost of one correction:



$$x^2 + 6x + 9 = (x + 3)^2$$

$$\begin{aligned} \text{so } x^2 + 6x \\ = (x + 3)^2 - 9 \end{aligned}$$

$x^2 + 6x$ is one small square short of $(x + 3)^2$. Add the missing 9 — and take it away again.

$$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$$

So $x^2 + 6x + 1 = (x + 3)^2 - 9 + 1 = (x + 3)^2 - 8$.

WHEN $A \neq 1$

Take a out of the first two terms first: $2x^2 + 8x + 3 = 2(x^2 + 4x) + 3 = 2[(x + 2)^2 - 4] + 3 = 2(x + 2)^2 - 5$. The bracket must be multiplied back out.

What the completed form gives you free

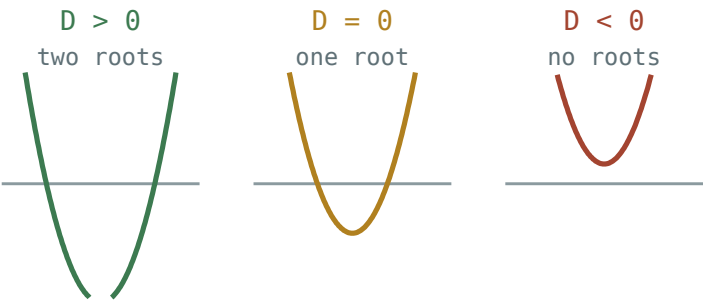
From $y = (x + p)^2 + q$ you can read, without any further work:

, at $x = -p$ — real only if $q \leq 0$	
QUESTION	ANSWER
the vertex	$(-p, q)$
the axis of symmetry	$x = -p$
the least value of y	q
the range	$y \geq q$
the roots	$(x + p)^2 = -q$

A square is never negative, so $(x + p)^2 \geq 0$ and therefore $y \geq q$ always. That one line answers every "find the minimum value" question in the quarter.

The discriminant

$$D = b^2 - 4ac$$



$D > 0$ cuts the axis twice, $D = 0$ touches it, $D < 0$ misses it entirely.

D	ROOTS	THE PARABOLA
$D > 0$	two different real roots	cuts the x -axis twice
$D = 0$	one repeated root	touches the axis — it is a tangent
$D < 0$	no real roots	misses the axis completely

“Touches the axis”, “has a repeated root”, “is a tangent to the x -axis” and “has exactly one solution” are four ways of writing $D = 0$. Recognising them is most of the marks in this topic.

02 Worked examples

EXAMPLE 1 Write $x^2 - 8x + 21$ in completed-square form and state its least value.

- 1 half of -8 is -4
- 2 $x^2 - 8x = (x - 4)^2 - 16$
- 3 $(x - 4)^2 - 16 + 21 = (x - 4)^2 + 5$
- 4 Least value 5, at $x = 4$.
A square is at least 0.

ANSWER

$(x - 4)^2 + 5$; least value 5

EXAMPLE 2 Write $3x^2 + 12x + 7$ in the form $a(x + p)^2 + q$.

① $3(x^2 + 4x) + 7$

Take the 3 out of the first two terms only.

② $x^2 + 4x = (x + 2)^2 - 4$

③ $3[(x + 2)^2 - 4] + 7$

④ $3(x + 2)^2 - 12 + 7 = 3(x + 2)^2 - 5$

Multiply the 3 through.

ANSWER

$$3(x + 2)^2 - 5$$

EXAMPLE 3 Find the values of k for which $x^2 + kx + 9 = 0$ has a repeated root.

① A repeated root means $D = 0$.

② $D = k^2 - 4 \cdot 1 \cdot 9 = k^2 - 36$

③ $k^2 - 36 = 0$

④ $k = \pm 6$

Both work: $(x \pm 3)^2$.

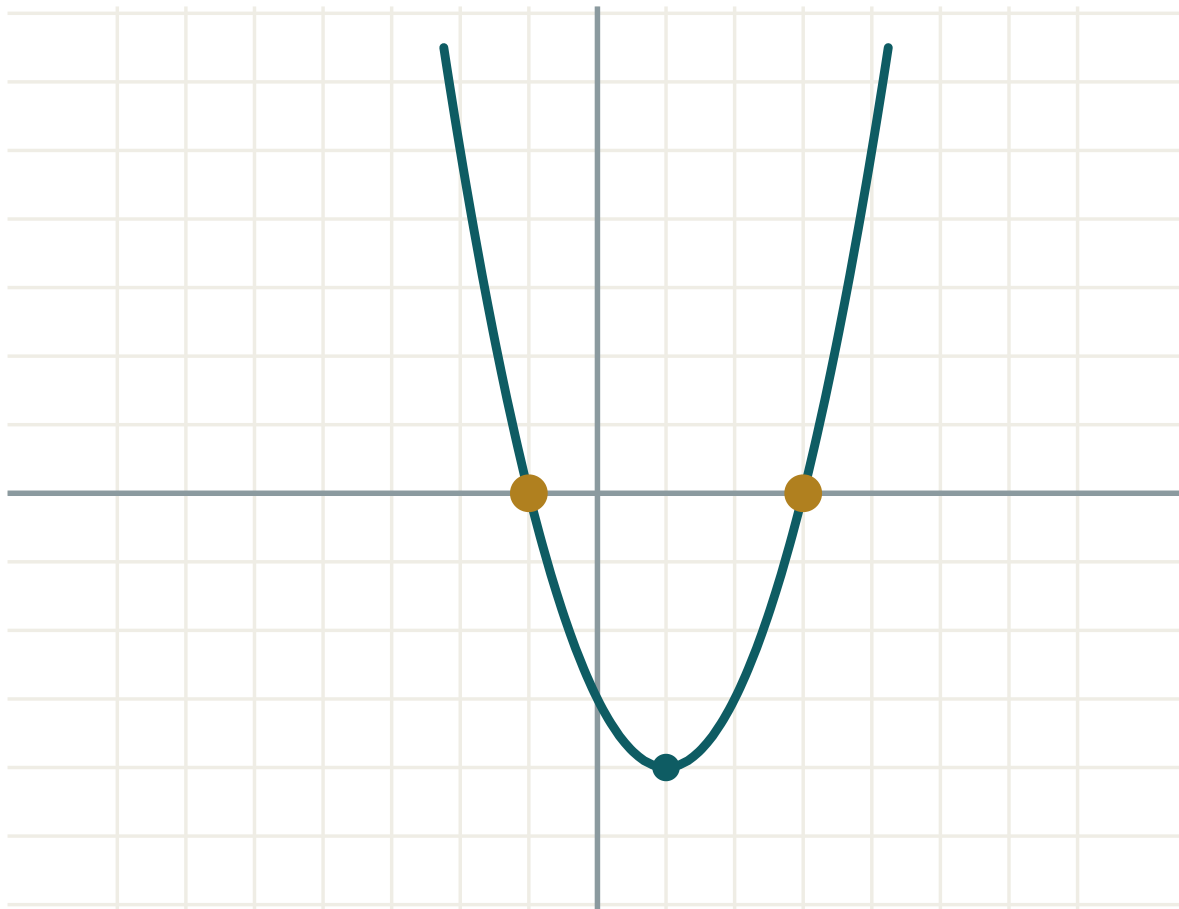
ANSWER

$$k = 6 \text{ or } k = -6$$

03 Interactive model

Move the coefficients and watch D change sign as the parabola lifts off the axis.

The discriminant in action



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Completing the square	To'la kvadratga keltirish	Выделение полного квадрата
Perfect square	To'la kvadrat	Полный квадрат
Vertex	Uchi	Вершина
Minimum value	Eng kichik qiymat	Наименьшее значение
Maximum value	Eng katta qiymat	Наибольшее значение
Discriminant	Diskriminant	Дискриминант
Repeated root	Karrali ildiz	Кратный корень
Tangent to the axis	O'qqa urinma	Касается оси
Axis of symmetry	Simmetriya o'qi	Ось симметрии

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$x^2 + 10x$ completed is:

The least value of $(x - 2)^2 + 7$ is:

If $D = 0$ the parabola:

The vertex of $y = (x + 4)^2 - 3$ is:

$(4, -3)$

$(-4, -3)$

$(-4, 3)$

$(4, 3)$

$x^2 + 2x + 5 = 0$ has:

two roots

one root

no real roots

infinitely many

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Complete the square: $x^2 + 4x$

ANSWER $(x + 2)^2 - 4$

2. Complete the square: $x^2 - 6x$

ANSWER $(x - 3)^2 - 9$

3. Complete the square: $x^2 + 2x + 5$

ANSWER $(x + 1)^2 + 4$

4. State the vertex of $y = (x - 1)^2 + 2$

ANSWER $(1, 2)$

5. Find D for $x^2 + 6x + 9 = 0$

ANSWER 0

6. Find D for $x^2 + x + 1 = 0$

ANSWER -3

7. State the least value of $(x + 3)^2 - 1$

ANSWER -1

Medium · the standard question

7 PROBLEMS

1. Complete the square: $x^2 - 10x + 30$

ANSWER $(x - 5)^2 + 5$

2. Complete the square: $2x^2 + 8x + 1$

ANSWER $2(x + 2)^2 - 7$

3. Find the vertex of $y = x^2 - 4x + 7$

ANSWER $(2, 3)$

4. Find k so that $x^2 + kx + 25 = 0$ has one root.

ANSWER $k = \pm 10$

5. For what k does $x^2 + 4x + k = 0$ have two roots?

ANSWER $k < 4$

6. Find the least value of $3x^2 - 12x + 5$

ANSWER $3(x - 2)^2 - 7$, least -7

7. Show $x^2 - 2x + 6$ is always positive.

ANSWER $(x - 1)^2 + 5 \geq 5 > 0$

Hard · stretch and reason

7 PROBLEMS

1. Complete the square: $5 - 4x - x^2$

ANSWER $9 - (x + 2)^2$, greatest value 9

2. For what k is $y = x^2 + kx + k + 3$ a tangent to the x -axis?

ANSWER $k^2 - 4k - 12 = 0$, so $k = 6$ or $k = -2$

3. Find the range of $y = 2x^2 - 8x + 11$

ANSWER $2(x - 2)^2 + 3$, so $y \geq 3$

4. The line $y = mx + 1$ is a tangent to $y = x^2 + 3$. Find m .

ANSWER $x^2 - mx + 2 = 0$, $D = m^2 - 8 = 0$, $m = \pm 2\sqrt{2}$

5. For what values of k does $kx^2 + 4x + 1 = 0$ have no real root?

ANSWER $16 - 4k < 0$, so $k > 4$

6. Prove $x^2 + bx + c$ has a least value of $c - b^2/4$.

ANSWER $(x + \frac{b}{2})^2 + c - \frac{b^2}{4}$; the square is at least 0.

7. Find the greatest value of $\frac{1}{x^2 - 4x + 8}$

ANSWER the denominator is least at 4, so the greatest value is $\frac{1}{4}$

07 Homework

Homework — 6 problems

P1 Exercise 1B and 1F. Always state the vertex once you have the completed form.

- Complete the square: $x^2 + 12x + 40$
- Complete the square: $2x^2 - 12x + 5$
- Find the vertex and least value of $y = x^2 - 6x + 11$
- Find k so that $x^2 + kx + 16 = 0$ has a repeated root.

5. *For what k does $x^2 + 6x + k = 0$ have no real root?*

6. *Show that $x^2 + 4x + 9$ is positive for every x .*

Quadratic inequalities and simultaneous equations

Cambridge insert: solving with a sketch rather than a rule, and the substitution that handles a curve meeting a line.

LESSON 6–7

2 × 40 MIN

ALGEBRA 10, REVISION

P1 · 1.2, 1.7

LESSON CLOCK



Sketch three parabolas and shade where $y > 0$ 6 min

Quadratic inequalities 14 min

Simultaneous equations 12 min

When does the line miss? 4 min

Homework 4 min

LEARNING OBJECTIVES

- Solve a quadratic inequality by sketching the parabola.
- Write the answer with the right connective — “and” or “or”.
- Solve one linear and one quadratic equation together.
- Use the discriminant to decide whether a line meets a curve.

TEXTBOOK

National Повторение, pp. 9–13

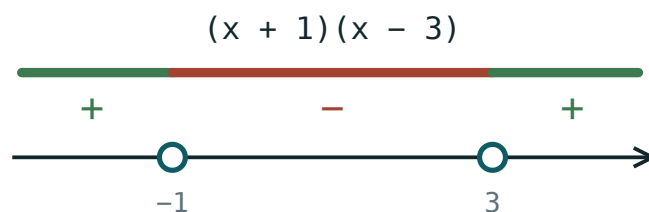
Cambridge Pure Mathematics 1, pp. 6, 16–17

Workbook P1 Exercise 1C, 16

01 Explanation

Inequalities: sketch, do not memorise

To solve $x^2 - 2x - 3 > 0$: find where the expression is **zero**, sketch the parabola, then read off where it is above the axis.



negative between the roots, positive outside

The critical values split the line into three. A parabola opening upwards is negative only between its roots.

THE METHOD, IN THREE STEPS

1. Make one side zero and factorise: $(x + 1)(x - 3) > 0$.
2. Mark the critical values -1 and 3 on a sketch of the parabola.

3. Read the answer off: above the axis **outside** the roots, so $x < -1$ or $x > 3$.

THE CONNECTIVE CARRIES A MARK

Between the roots the answer is **one** interval and takes “and”: $-1 < x < 3$. Outside the roots it is **two** intervals and takes “or”: $x < -1$ or $x > 3$. Writing $3 < x < -1$ says nothing exists.

One linear and one quadratic

Always substitute from the **linear** equation into the quadratic — never the other way round. Solve $y = x + 1$ with $x^2 + y^2 = 25$:

$$x^2 + (x + 1)^2 = 25 \implies 2x^2 + 2x - 24 = 0 \implies x^2 + x - 12 = 0$$

So $x = 3$ or $x = -4$, and the line $y = x + 1$ gives $y = 4$ and $y = -3$. Two intersection points: $(3, 4)$ and $(-4, -3)$.

PAIR THE ANSWERS UP

Each x belongs with **its own** y . Listing $x = 3, -4$ and $y = 4, -3$ separately loses the pairing, and the mark.

Does the line meet the curve at all?

After the substitution you have a quadratic. Its discriminant answers the geometry:

D	THE LINE AND THE CURVE
$D > 0$	meet at two points
$D = 0$	touch — the line is a tangent
$D < 0$	never meet

So “find m so that $y = mx + 3$ is a tangent to $y = x^2$ ” is not a geometry question at all.

Substitute, set $D = 0$, solve.

02 Worked examples

EXAMPLE 1 Solve $x^2 - x - 6 \leq 0$.

① $(x - 3)(x + 2) \leq 0$

Factorise.

② *critical values* -2 and 3

③ The parabola opens upwards, so it is at or below the axis **between** the roots.

④ $-2 \leq x \leq 3$

Closed circles: $\leq$ includes the ends.

ANSWER

$$-2 \leq x \leq 3$$

EXAMPLE 2 Solve $y = 2x - 1$ together with $y = x^2 - 4$.

① $2x - 1 = x^2 - 4$

Both equal y .

② $x^2 - 2x - 3 = 0$

③ $(x - 3)(x + 1) = 0$

④ $x = 3 \Rightarrow y = 5; x = -1 \Rightarrow y = -3$

Pair them.

ANSWER

$$(3, 5) \text{ and } (-1, -3)$$

EXAMPLE 3 Find c so that $y = 3x + c$ is a tangent to $y = x^2 + 5$.

① $3x + c = x^2 + 5$

② $x^2 - 3x + 5 - c = 0$

③ $D = 9 - 4(5 - c) = 0$

Tangent means one repeated root.

④ $9 - 20 + 4c = 0$

⑤ $c = \frac{11}{4}$

ANSWER

$$c = \frac{11}{4}$$

03 Interactive model

Move the coefficients; watch which region of the number line makes the parabola positive.

Where is the expression positive?

$$x > 2$$



inequality $x > 2$

boundary 2 (open)

$x = 0$ works? no

$x = 6$ works? yes

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Quadratic inequality	Kvadrat tengsizlik	Квадратное неравенство
Critical value	Kritik qiymat	Критическое значение
Sign chart	Ishoralar jadvali	Схема знаков
Sketch	Eskiz	Эскиз
Simultaneous equations	Tenglamalar sistemasi	Система уравнений
Substitution method	O‘rniga qo‘yish usuli	Метод подстановки
Point of intersection	Kesishish nuqtasi	Точка пересечения
Tangent	Urinma	Касательная
Interval	Oraliq	Промежуток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$(x - 1)(x - 4) < 0$ gives:

$x^2 > 9$ gives:

Substituting a line into a curve gives $D < 0$. They:

Solving $y = x$ with $y = x^2$ gives:

$(0,0)$ only

$(1,1)$ only

$(0,0)$ and $(1,1)$

no points

Which is impossible?

$2 < x < 5$

$x < 2$ or $x > 5$

$5 < x < 2$

$x \geq 5$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $(x - 2)(x - 5) < 0$

ANSWER $2 < x < 5$

2. Solve $(x + 1)(x - 3) > 0$

ANSWER $x < -1$ or $x > 3$

3. Solve $x^2 < 16$

ANSWER $-4 < x < 4$

4. Solve $x^2 \geq 25$

ANSWER $x \leq -5$ or $x \geq 5$

5. Solve $x^2 - 4x \leq 0$

ANSWER $0 \leq x \leq 4$

6. Solve $y = x + 2$ and $y = 2x$

ANSWER $(2, 4)$

7. Solve $y = x^2$ and $y = 4$

ANSWER $(\pm 2, 4)$

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 - 5x + 6 > 0$

ANSWER $x < 2$ or $x > 3$

2. Solve $2x^2 + x - 6 \leq 0$

ANSWER $-2 \leq x \leq 1.5$

3. Solve $6 - x - x^2 > 0$

ANSWER $-3 < x < 2$

4. Solve $y = x - 1$ and $y = x^2 - 3$

ANSWER $(2, 1)$ and $(-1, -2)$

5. Solve $y = 2x$ and $x^2 + y^2 = 20$

ANSWER $(2, 4)$ and $(-2, -4)$

6. Find c so $y = 4x + c$ is a tangent to $y = x^2$.

ANSWER $c = -4$

7. Solve $x^2 + 3x \geq 10$

ANSWER $x \leq -5$ or $x \geq 2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $x^2 - 2x + 5 > 0$

ANSWER every x — $D < 0$ and the parabola opens upwards

2. Solve $x^2 + 4x + 4 \leq 0$

ANSWER $x = -2$ only

3. Solve $\frac{x-1}{x+2} > 0$

ANSWER $x < -2$ or $x > 1$

4. Find m so $y = mx - 2$ is a tangent to $y = x^2 + 2$.

ANSWER $x^2 - mx + 4 = 0$, $m^2 = 16$, $m = \pm 4$

5. Solve $y = 3 - x$ and $x^2 + y^2 = 5$

ANSWER $(1, 2)$ and $(2, 1)$

6. For what k does $y = x + k$ never meet $y = x^2$?

ANSWER $x^2 - x - k = 0$, $1 + 4k < 0$, so $k < -\frac{1}{4}$

7. Solve $x^4 - 5x^2 + 4 < 0$

ANSWER $t = x^2$: $1 < t < 4$, so $1 < |x| < 2$ — that is $-2 < x < -1$ or $1 < x < 2$

07 Homework

Homework — 6 problems

P1 Exercise 1C and 1G. Sketch the parabola for every inequality.

1. Solve $(x - 1)(x + 4) < 0$

2. Solve $x^2 - 7x + 10 \geq 0$

3. Solve $3 + 2x - x^2 > 0$

4. Solve $y = 2x + 1$ and $y = x^2 - 2$

5. Solve $y = x - 2$ and $x^2 + y^2 = 10$
6. Find c so that $y = 2x + c$ is a tangent to $y = x^2 + 1$.

Mappings — project lesson

One hour, worked in pairs: build a mapping, decide whether it is a function, and defend the decision.

LESSON 8

1 × 40 MIN

ALGEBRA 10, §1.1

P1 · 2.1

LESSON CLOCK

5'

8'

20'

10'

2

What a mapping is	5 min
The four kinds	8 min
Pair work — build and classify	20 min
Two pairs present	10 min
Homework	2 min

LEARNING OBJECTIVES

- Describe a mapping by an arrow diagram, a rule and a set of ordered pairs.
- Classify a mapping as one-to-one, many-to-one, one-to-many or many-to-many.
- Decide whether a mapping is a function and justify the decision.
- Present a mapping the pair invented and answer questions about it.

TEXTBOOK

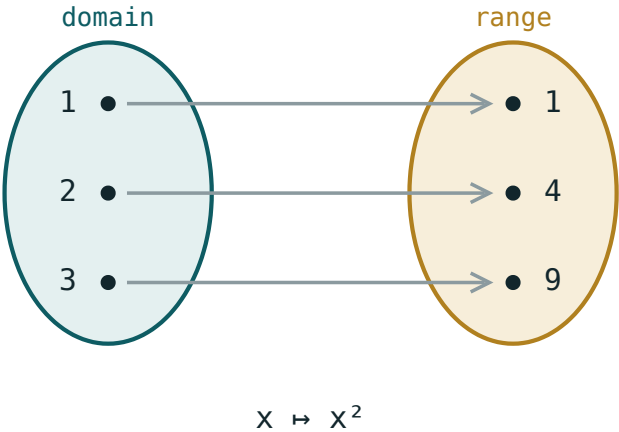
National	pp. 24–27
Cambridge	Pure Mathematics 1, pp. 26–29
Workbook	Project sheet M1

01

Explanation

A mapping is a rule with a direction

Take a set of inputs and a rule. Apply the rule to each input and you get an output. That is a **mapping**. Nothing more is required — the rule does not have to be a formula, and the sets do not have to be numbers.



Four mappings. Read each one left to right and count the arrows leaving, then arriving at, every dot.

ONLY TWO QUESTIONS MATTER

How many arrows leave each input? How many arrive at each output? The first answer decides whether the mapping is a function. The second decides whether it can be undone.

The four kinds

KIND	ARROWS OUT OF EACH INPUT	ARROWS INTO AN OUTPUT	EXAMPLE
one-to-one	exactly one	at most one	$x \mapsto 2x + 1$
many-to-one	exactly one	more than one, somewhere	$x \mapsto x^2$
one-to-many	more than one, somewhere	at most one	$x \mapsto \pm\sqrt{x}$
many-to-many	more than one	more than one	student $\mapsto$ subject studied

THE DEFINITION OF A FUNCTION

A mapping is a **function** when every input has **exactly one** output. So the first two rows are functions and the last two are not.

THE COMMONEST ERROR

$x \mapsto x^2$ sends 3 and -3 both to 9. Two arrows arrive at 9 — but only one leaves each input, so it **is** a function. Arrows arriving never break the definition.

The project

In pairs, twenty minutes. Produce one A4 sheet carrying all four of these:

1. A mapping of your own, on a set of at least six inputs, that is **not** numerical.
2. The same mapping as an arrow diagram **and** as a set of ordered pairs.
3. Its classification, with one sentence of justification for each half of the name.
4. One small change to the mapping that turns it into a different kind, and the new name.

MARKING, OUT OF 8

2 for the mapping being genuinely your own · 2 for both representations agreeing · 2 for a correct classification · 2 for the change in part 4 actually working.

Two pairs present in the last ten minutes. The class asks one question each: *“Is it a function, and how do you know from your diagram alone?”*

02 Worked examples

EXAMPLE 1 Classify the mapping $x \mapsto x^3$ on the real numbers.

- 1 Each x has one cube.
One arrow leaves each input.
- 2 Can two inputs share a cube?
 $a^3 = b^3 \Rightarrow a = b$ for real numbers — no.
- 3 So at most one arrow arrives anywhere.

ANSWER

One-to-one, and therefore a function

EXAMPLE 2 Is $x \mapsto \sqrt{x}$ with output “any square root of x ” a function?

- ① $x = 9$ maps to 3 and to -3 .
Two arrows leave one input.

- ② The definition demands exactly one.

ANSWER

No — it is one-to-many, not a function

EXAMPLE 3 The mapping “capital city of” on {Uzbekistan, France, Japan}. Classify it.

- ① Each country has exactly one capital.
A function.

- ② No two of these countries share a capital.
At most one arrow arrives.

ANSWER

One-to-one

03 Interactive model

Drag the inputs to see when a mapping stops being a function.

Function or not? $x \mapsto 2x - 7$ is:☐ one-to-one☐ many-to-one☐ one-to-many☐ not a mapping $x \mapsto |x|$ is:☐ one-to-one☐ many-to-one☐ one-to-many☐ many-to-manyWhich of these is **not** a function?☐ $x \mapsto x^2$ ☐ $x \mapsto x^3$ ☐ $x^2 + y^2 = 25$ ☐ $x \mapsto 5$

$x \mapsto 5$ for every x is:

not a function

one-to-one

many-to-one

one-to-many

A mapping can be undone by a mapping when it is:

many-to-one

one-to-one

one-to-many

any function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Mapping	Akslantirish	Отображение
Object (input)	Argument	Аргумент
Image (output)	Tasvir	Образ
One-to-one	Biror-birga	Взаимно однозначное
Many-to-one	Ko'pdan-birga	Многие в одно
One-to-many	Birdan-ko'pga	Одно во многие
Arrow diagram	Strelkali diagramma	Стрелочная диаграмма
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
Function	Funksiya	Функция

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A function is a mapping in which every input has:

$x \mapsto x^2$ on the reals is:

Ordered pairs $\{(1,2), (2,4), (1,5)\}$ describe:

The set of all first coordinates is the:

range

domain

image

codomain

06

Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Classify $x \mapsto x + 4$

ANSWER one-to-one

2. Classify $x \mapsto 3$

ANSWER many-to-one

3. Classify $x \mapsto \pm x$

ANSWER one-to-many

4. Is $\{(1,1), (2,4), (3,9)\}$ a function?

ANSWER yes

5. Is $\{(1,1), (1,2)\}$ a function?

ANSWER no

6. Classify $x \mapsto x^4$ on the reals

ANSWER many-to-one

7. Name the domain of $\{(0,5), (2,7)\}$

ANSWER $\{0, 2\}$

Medium · the standard question

7 PROBLEMS

1. Classify $x \mapsto 2^x$

ANSWER one-to-one

2. Classify $x \mapsto \sin x$ on the reals

ANSWER many-to-one

3. Is $x^2 + y^2 = 9$ a function of x ?

ANSWER no

4. Give a mapping on $\{a, b, c\}$ that is many-to-one

ANSWER e.g. all three $\mapsto 1$

5. Write $x \mapsto 3x - 1$ on $\{0, 1, 2\}$ as ordered pairs

ANSWER $\{(0, -1), (1, 2), (2, 5)\}$

6. Classify “mother of” on a set of people

ANSWER many-to-one

7. Classify “sibling of”

ANSWER many-to-many

Hard · stretch and reason

7 PROBLEMS

1. Restrict $x \mapsto x^2$ to make it one-to-one

ANSWER $x \geq 0$ (or $x \leq 0$)

2. Explain why one-to-many mappings can never be functions

ANSWER One input would need two images

3. Is $x \mapsto \frac{1}{x}$ one-to-one on $x \neq 0$?

ANSWER yes

4. Find all k making $x \mapsto x^2 + kx$ one-to-one on $x \geq 0$

ANSWER $k \geq 0$

5. How many functions map $\{1, 2\}$ to $\{a, b\}$?

ANSWER 4

6. How many of those are one-to-one?

ANSWER 2

7. A mapping on a finite set is one-to-one exactly when it is what?

ANSWER onto — it uses every output

07 Homework

Homework — 4 tasks

Finish the project sheet if your pair did not.

1. Complete the four parts of the project sheet.
2. Classify: $x \mapsto x^5$, $x \mapsto x^6$, $x \mapsto |x| + 1$.
3. Write a mapping from your own life that is many-to-many.
4. Explain in two sentences why every one-to-one mapping is a function but not every function is one-to-one.

The function and the ways of defining it

The same function written five ways — formula, table, graph, words and a machine — and why a question sometimes wants one rather than another.

LESSON 9–10

2 × 40 MIN

ALGEBRA 10, §1.2

P1 · 2.1–2.2

LESSON CLOCK

6'

12'

8'

12'

7'

5'

Recall — exactly one output	6 min
Five ways of writing one function	12 min
Function notation	8 min
The vertical line test	12 min
Practice	7 min
Homework	5 min

LEARNING OBJECTIVES

- State the definition of a function in terms of domain, rule and image.
- Move between formula, table, graph and verbal description of the same function.
- Use function notation $f(x)$ and evaluate f at a number and at an expression.
- Apply the vertical line test to a graph.

TEXTBOOK

National	pp. 28–34
Cambridge	Pure Mathematics 1, pp. 26–33
Workbook	P1 Exercise 2A

01

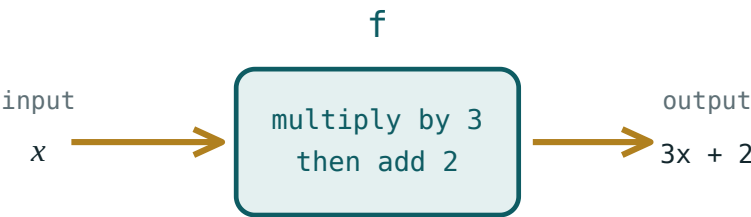
Explanation

What has to be given

A FUNCTION NEEDS THREE THINGS

A **domain** — the set of allowed inputs. A **rule** — what to do to an input. And the guarantee that the rule gives **exactly one** output for each input.

Change the domain and you change the function, even if the rule is unchanged. $f(x) = x^2$ on $\mathbb{R}$ is many-to-one; the same rule on $x \geq 0$ is one-to-one. They are different functions.



The machine picture: one input in, the rule applied, exactly one output out.

Five ways to write the same function

The doubling-and-adding-one function, five ways:

WAY	HOW IT LOOKS	BEST WHEN
formula	$f(x) = 2x + 1$	you must compute or manipulate
table	$0 \mapsto 1, 1 \mapsto 3, 2 \mapsto 5$	the domain is small or the data is measured
graph	a straight line, gradient 2	you want shape, roots, behaviour
words	“double the number, then add one”	explaining, or modelling a real situation
ordered pairs	$\{(0,1), (1,3), (2,5)\}$	the definition must be checked exactly

A TABLE IS NOT ALWAYS ENOUGH

Three points do not determine a function. $f(x) = 2x + 1$ and a cubic through the same three points agree on the table and disagree everywhere else. A table defines a function only when the domain is exactly the listed inputs.

Function notation

$f(3)$ means “the output when the input is 3”. It is not multiplication. The letter inside is a placeholder, so it can be replaced by anything — a number, another letter, a whole expression.

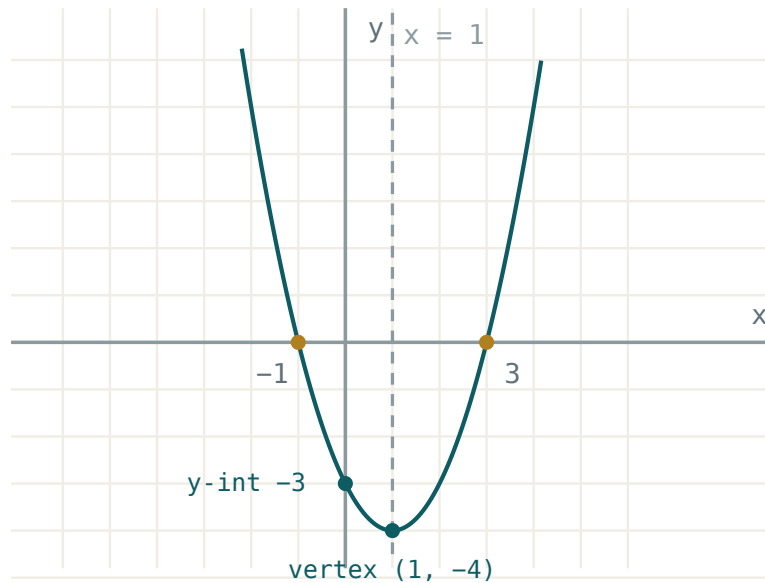
$$f(x) = x^2 - 3x \Rightarrow f(2) = -2, f(a) = a^2 - 3a, f(x + 1) = (x + 1)^2 - 3(x + 1)$$

SUBSTITUTE THE WHOLE THING

$f(x + 1)$ is **not** $f(x) + 1$. Put brackets round the substituted expression every time, then expand. Most lost marks in this topic live in that one missing bracket.

The vertical line test

A graph shows a function exactly when no vertical line meets it more than once — that is the picture of “exactly one output per input”.



Every vertical line meets this parabola once. It is a function — many-to-one, but a function.

A circle fails: the line $x = 0$ meets $x^2 + y^2 = 25$ at $(0, 5)$ and $(0, -5)$. So a circle is a relation, not a function of x .

02 Worked examples

EXAMPLE 1 For $f(x) = x^2 - 4x + 1$, find $f(0)$, $f(-2)$ and $f(a + 1)$.

① $f(0) = 0 - 0 + 1 = 1$

② $f(-2) = 4 + 8 + 1 = 13$
Watch $-4 \cdot (-2) = +8$.

③ $f(a + 1) = (a + 1)^2 - 4(a + 1) + 1$
Brackets first.

④ $= a^2 + 2a + 1 - 4a - 4 + 1 = a^2 - 2a - 2$

ANSWER

$1, 13, a^2 - 2a - 2$

EXAMPLE 2 If $g(x) = 3x - 5$ and $g(k) = 7$, find k .

① $3k - 5 = 7$
Read the notation as an equation.

② $3k = 12$

③ $k = 4$

ANSWER

$k = 4$

EXAMPLE**3**

A taxi charges 8000 so‘m plus 2500 so‘m a kilometre. Write the cost as a function and find the cost of 12 km.

1

Words to formula.

$$C(d) = 8000 + 2500d$$

2

State the domain.

$$d \geq 0$$

3

$$C(12) = 8000 + 30000 = 38000$$

ANSWER

$$C(d) = 8000 + 2500d; 38\,000 \text{ so‘m}$$

03 Interactive model

Change the rule and watch the table, the graph and the machine all update together.

Reading function notation

If $f(x) = 5 - 2x$ then $f(4)$ is:

If $f(x) = x^2$ then $f(x + 2)$ is:

A vertical line cutting a graph twice means:

If $h(x) = \underline{1} x - 3$ then $h(3)$ is:

0

1

undefined

3

$f(x) = 4$ for all x has a graph that is:

a vertical line

a horizontal line

a parabola

not a graph

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Function	Funksiya	Функция
Function notation	Funksional belgilash	Функциональное обозначение
Independent variable	Erkli o'zgaruvchi	Независимая переменная
Dependent variable	Erksiz o'zgaruvchi	Зависимая переменная
Rule	Qoida	Правило
Table of values	Qiyimatlar jadvali	Таблица значений
Graph of a function	Funksiya grafigi	График функции
Vertical line test	Vertikal chiziq sinovi	Тест вертикальной прямой
Piecewise definition	Bo'lakli berilish	Кусочное задание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$f(x) = 2x + 3$, so $f(-1)$ is:

The independent variable in $y = f(x)$ is:

Which cannot be the graph of a function?

If $f(x) = x^2$, then $f(a) = f(b)$ means:

$$a = b$$

$$a = \pm b$$

$$a = -b$$

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $f(x) = 3x + 1$; find $f(2)$

ANSWER 7

2. $f(x) = 3x + 1$; find $f(0)$

ANSWER 1

3. $f(x) = x^2 - 1$; find $f(-3)$

ANSWER 8

4. $g(x) = 10 - x$; find $g(10)$

ANSWER 0

5. Is a circle the graph of a function?

ANSWER no

6. $f(x) = 7$; find $f(100)$

ANSWER 7

7. Write “treble it, then subtract 2” as a formula

ANSWER $f(x) = 3x - 2$

Medium · the standard question

7 PROBLEMS

1. $f(x) = x^2 - 2x$; find $f(a + 1)$

ANSWER $a^2 - 1$

2. $f(x) = 4x - 9$; solve $f(x) = 0$

ANSWER $x = 2.25$

3. $f(x) = \frac{x}{x + 2}$; find $f(2)$

ANSWER 0.5

4. $f(x) = \frac{x}{x + 2}$; which input is forbidden?

ANSWER $x = -2$

5. $f(x) = x^3$; solve $f(x) = 8$

ANSWER $x = 2$

6. A phone plan costs 30 000 plus 200 a minute. Write $C(t)$

ANSWER $C(t) = 30000 + 200t$

7. $f(x) = 2x$; show $f(a + b) = f(a) + f(b)$

ANSWER $2(a + b) = 2a + 2b$ ✓

Hard · stretch and reason

7 PROBLEMS

1. $f(x) = x^2 + 1$; simplify $\frac{f(x+h) - f(x)}{h}$

ANSWER $2x + h$

2. $f(x) = \frac{1}{x}$; simplify $f(a) - f(b)$

ANSWER $\frac{b-a}{ab}$

3. $f(x) = 2x + 3$; find x with $f(x) = f(2x)$

ANSWER $x = 0$

4. $f(x) = x^2$; find all a with $f(a) = f(a+2)$

ANSWER $a = -1$

5. Give a rule whose graph fails the vertical line test

ANSWER $x = y^2$

6. $f(x) = ax + b$, $f(1) = 5$, $f(3) = 11$. Find a , b

ANSWER $a = 3$, $b = 2$

7. Explain why $f(x+1) \neq f(x) + 1$ in general

ANSWER Only additive rules satisfy it — e.g. $f(x) = x^2$ fails

67 Homework

Homework — 5 tasks

Task 5 needs the vertical line test drawn, not just named.

- $f(x) = x^2 - 5x + 6$. Find $f(0)$, $f(2)$, $f(-1)$.
- $f(x) = 3 - 4x$. Solve $f(x) = 15$.
- $f(x) = x^2 + x$. Simplify $f(x+1) - f(x)$.
- Write the same function four ways: formula, table on $\{0,1,2,3\}$, words and ordered pairs.

5. Sketch a curve that is a function and one that is not, and mark the vertical line that decides each.

Domain and range

Two questions, asked in a fixed order: what may go in, and what then comes out.

LESSON 11–12

2 × 40 MIN

ALGEBRA 10, §1.3

P1 · 2.2

LESSON CLOCK

6'

14'

14'

12'

8'

6'

The two questions	6 min
Finding a domain — three traps	14 min
Finding a range	14 min
Restricted domains	12 min
Practice	8 min
Homework	6 min

LEARNING OBJECTIVES

- Find the natural domain of a rational, root and polynomial function.
- Find the range of a linear, quadratic and simple rational function.
- Write domain and range in set notation and in interval notation.
- Explain the effect of a restricted domain on the range.

TEXTBOOK

National	pp. 35–41
Cambridge	Pure Mathematics 1, pp. 30–36
Workbook	P1 Exercise 2B

01

Explanation

Domain first, always

THE ORDER IS NOT OPTIONAL

The **domain** is the set of inputs the rule accepts. The **range** is the set of outputs it then produces. You cannot know the range until you know which inputs were allowed — so find the domain first, every time.

When a function is given by a formula and nothing else, its domain is the largest set of real numbers the formula makes sense on. That is the **natural domain**.

Three things a formula can refuse

REFUSAL	CONDITION TO IMPOSE	EXAMPLE	DOMAIN
division by zero	denominator $\neq 0$	$\frac{1}{x-3}$	$x \neq 3$
even root of a negative	radicand ≥ 0	$\sqrt{x-2}$	$x \geq 2$
both at once	radicand > 0	$\frac{1}{\sqrt{x-2}}$	$x > 2$

A polynomial refuses nothing: its natural domain is always $\mathbb{R}$.

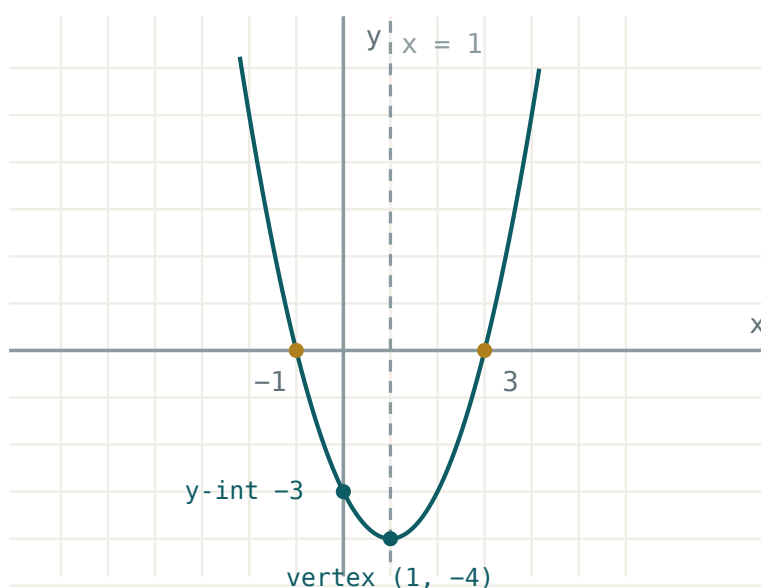
CANCEL AFTER THE DOMAIN, NOT BEFORE

$f(x) = \frac{x^2 - 9}{x - 3}$ simplifies to $x + 3$, but the domain is still $x \neq 3$. The graph is the line with a hole punched at $(3, 6)$. Cancelling never restores a forbidden input.

Finding the range

Four reliable methods, in the order you should try them:

1. **Linear** — the range is all of $\mathbb{R}$ unless the domain is restricted.
2. **Quadratic** — complete the square; the vertex gives the least (or greatest) value.
3. **Rational** — look for the value the output can never take: the horizontal asymptote.
4. **Anything** — sketch it and read the y -values the curve covers.



The range of a parabola starts at the vertex. Read it off the y -axis, not the x -axis.

$$y = x^2 - 6x + 11 = (x - 3)^2 + 2 \Rightarrow \text{range } y \geq 2$$

For $f(x) = \frac{1}{x}$ the output 0 is impossible, because $1 \div anything$ is never 0 . So the range is $y \neq 0$ — the same shape of answer as the domain, one value removed.

Restricting the domain

Give a function a smaller domain and the range shrinks with it. For $f(x) = x^2$:

DOMAIN	RANGE	KIND
$\mathbb{R}$	$y \geq 0$	many-to-one
$x \geq 0$	$y \geq 0$	one-to-one
$1 \leq x \leq 3$	$1 \leq y \leq 9$	one-to-one
$-1 \leq x \leq 3$	$0 \leq y \leq 9$	many-to-one

THE LAST ROW IS THE TRAP

On $-1 \leq x \leq 3$ the smallest output is **not** $f(-1) = 1$. The vertex $x = 0$ lies inside the interval, so the least value is 0 . Always check whether the vertex is inside the restricted domain before reading the endpoints.

02

Worked examples

EXAMPLE 1 Find the domain and range of $f(x) = \sqrt{x + 4}$.

1

Radicand ≥ 0 .
 $x + 4 \geq 0 \Rightarrow x \geq -4$

2

A square root gives non-negative outputs.
 $f(x) \geq 0$

3

At $x = -4$ the output is 0 and it grows without bound.

ANSWER

domain $x \geq -4$, range $y \geq 0$

EXAMPLE 2 Find the domain and range of $f(x) = \frac{2x}{x-1}$.

① Denominator $\neq 0$.

$$x \neq 1$$

② Rearrange for x : $y(x-1) = 2x$

$$yx - y = 2x$$

③ $x(y-2) = y \Rightarrow x = \frac{y}{y-2}$

This fails when $y = 2$.

④ So 2 is the one unreachable output.

ANSWER

domain $x \neq 1$, range $y \neq 2$

EXAMPLE 3 Find the range of $f(x) = x^2 - 4x + 7$ on $0 \leq x \leq 3$.

① Complete the square.

$$(x-2)^2 + 3$$

② The vertex $x = 2$ lies inside $[0, 3]$.

Least value 3.

③ Test both ends.

$$f(0) = 7, f(3) = 4$$

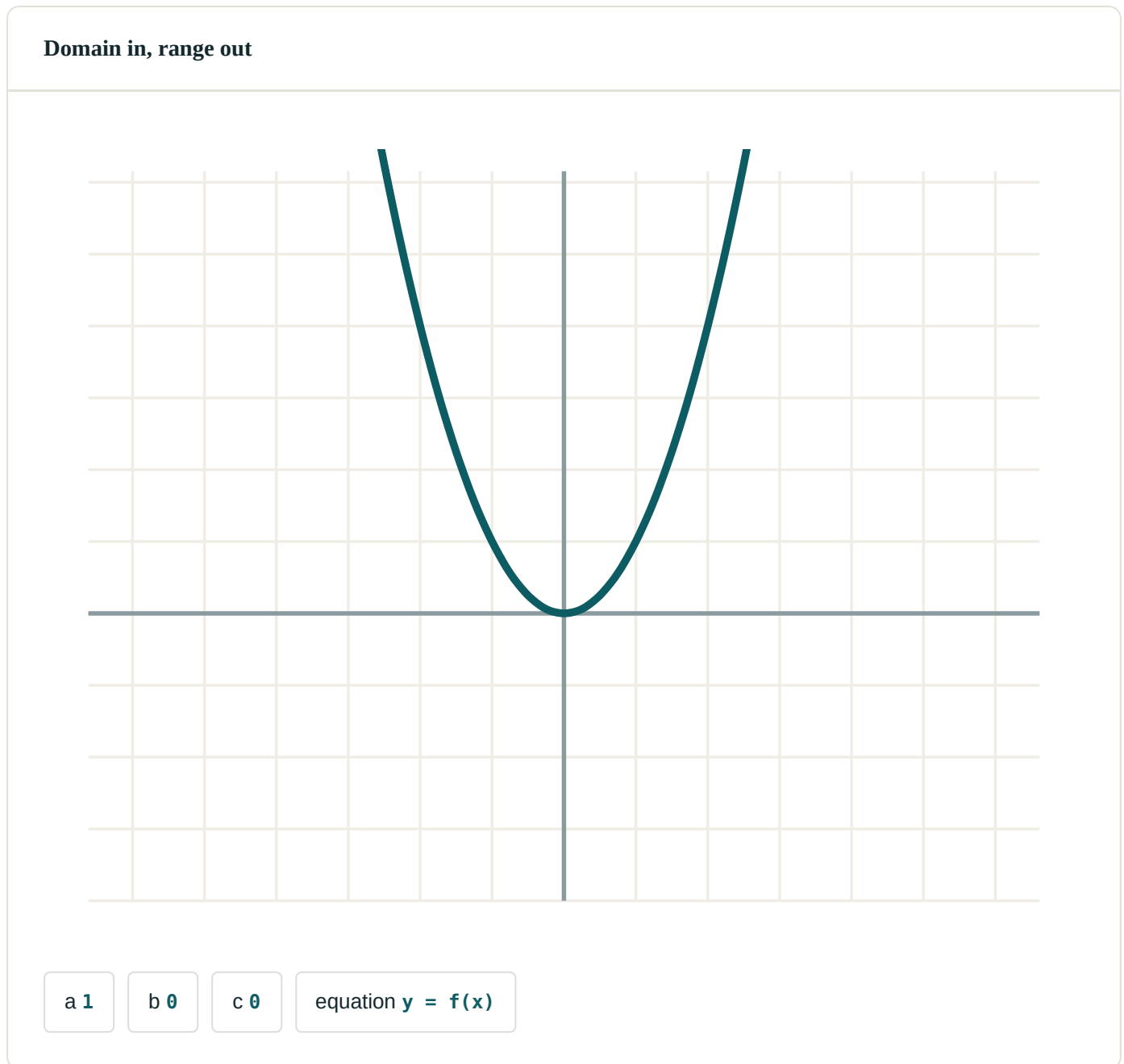
④ Greatest is 7.

ANSWER

$$3 \leq y \leq 7$$

03 Interactive model

Drag the ends of the domain and watch the range respond.



04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Domain	Aniqlanish sohasi	Область определения
Range	Qiymatlar sohasi	Область значений
Natural domain	Tabiiy aniqlanish sohasi	Естественная область определения
Interval notation	Oraliq belgilanishi	Интервальная запись
Excluded value	Chetlatilgan qiymat	Исключённое значение
Restriction	Cheklash	Ограничение
Asymptote	Asimptota	Асимптота
Minimum value	Eng kichik qiymat	Наименьшее значение
Set notation	To'plam belgilanishi	Обозначение множеств

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The natural domain of $\frac{1}{x+5}$ is:

$x \neq 5$

$x \neq -5$

$x \geq -5$

$\mathbb{R}$

The range of $y = x^2 + 3$ is:

$y \geq 0$

$y \geq 3$

$y > 3$

$\mathbb{R}$

The domain of $\sqrt{5-x}$ is:

$x \geq 5$

$x \leq 5$

$x \neq 5$

$\mathbb{R}$

The range of any non-constant linear function on $\mathbb{R}$ is:

$$y \geq 0$$

$$\mathbb{R}$$

$$y \neq 0$$

one value

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $f(x) = 3x - 1$

ANSWER $\mathbb{R}$

2. Domain of $\frac{1}{x}$

ANSWER $x \neq 0$

3. Domain of $\sqrt{x}$

ANSWER $x \geq 0$

4. Range of $y = x^2$

ANSWER $y \geq 0$

5. Range of $y = x^2 + 5$

ANSWER $y \geq 5$

6. Domain of $\frac{1}{x - 7}$

ANSWER $x \neq 7$

7. Range of $y = 4$

ANSWER $\{4\}$

Medium · the standard question

7 PROBLEMS

1. Domain of $\sqrt{2x - 6}$

ANSWER $x \geq 3$

2. Domain of $\frac{x}{x^2 - 4}$

ANSWER $x \neq \pm 2$

3. Range of $y = (x - 1)^2 - 6$

ANSWER $y \geq -6$

4. Range of $y = 7 - x^2$

ANSWER $y \leq 7$

5. Range of $y = \frac{1}{x} + 3$

ANSWER $y \neq 3$

6. Range of $y = 2x + 1$ on $0 \leq x \leq 4$

ANSWER $1 \leq y \leq 9$

7. Domain of $\frac{1}{\sqrt{x - 1}}$

ANSWER $x > 1$
Hard · stretch and reason

7 PROBLEMS

1. Range of $y = x^2 - 6x + 5$ on $0 \leq x \leq 2$

ANSWER $-3 \leq y \leq 5$

2. Range of $y = x^2 - 6x + 5$ on $1 \leq x \leq 5$

ANSWER $-4 \leq y \leq 0$

3. Range of $y = \frac{3x + 1}{x - 2}$

ANSWER $y \neq 3$

4. Domain of $\sqrt{x^2 - 9}$

ANSWER $x \leq -3$ or $x \geq 3$

5. Domain of $\sqrt{6 + x - x^2}$

ANSWER $-2 \leq x \leq 3$

6. Find k so $y = x^2 + kx + 10$ has range $y \geq 1$

ANSWER $k = \pm 6$

7. Why does $\frac{x^2 - 1}{x - 1}$ have range $y \neq 2$?

ANSWER It is $x + 1$ with the point at $x = 1$ removed

07 Homework

Homework — 5 tasks

Every answer must say *domain* and *range* separately, in that order.

- Domain and range of $f(x) = \sqrt{x - 3}$.
- Domain and range of $f(x) = \frac{5}{x + 2}$.
- Range of $f(x) = x^2 - 8x + 20$ on $\mathbb{R}$, then on $0 \leq x \leq 3$.

4. Explain why $\frac{x^2 - 4}{x + 2}$ does not have domain $\mathbb{R}$.
5. Invent a function with domain $x > 1$ and range $y > 0$.

Arithmetic operations on functions

Adding, subtracting, multiplying and dividing functions — and the one rule about the domain that decides every answer.

LESSON 13–14

2 × 40 MIN

ALGEBRA 10, §1.4

P1 · 2.3

LESSON CLOCK

6'

12'

10'

14'

8'

5'

Four new functions from two old ones	6 min
The domain rule	12 min
Division — the extra condition	10 min
Worked practice	14 min
Consolidation	8 min
Homework	5 min

LEARNING OBJECTIVES

- Form $f + g$, $f - g$, $f \cdot g$ and f/g from two given functions.
- Find the domain of a combined function as the intersection of the domains.
- Exclude the zeros of the denominator when dividing.
- Evaluate a combined function at a point.

TEXTBOOK

National	pp. 42–47
Cambridge	Pure Mathematics 1, pp. 37–41
Workbook	P1 Exercise 2C

01

Explanation

Four new functions from two

Given f and g , four new functions are defined pointwise — apply both rules to the same input, then combine the two outputs:

$$(f + g)(x) = f(x) + g(x) \cdot (f - g)(x) = f(x) - g(x)$$

$$(f \cdot g)(x) = f(x) \cdot g(x) \cdot \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

Nothing new is happening to the arithmetic. Everything new is happening to the domain.

The domain rule

BOTH RULES MUST ACCEPT THE INPUT

An input is allowed in $f + g$ only if it is allowed in f **and** in g . So the domain of the combination is the **intersection** of the two domains.

With $f(x) = \sqrt{x}$ (domain $x \geq 0$) and $g(x) = \frac{1}{x - 4}$ (domain $x \neq 4$), the sum has domain $x \geq 0$ and $x \neq 4$.

SIMPLIFYING DOES NOT WIDEN A DOMAIN
 $f(x) = \sqrt{x}$ and $g(x) = -\sqrt{x}$ give $(f + g)(x) = 0$ — but only for $x \geq 0$. The zero function on $\mathbb{R}$ is a different function.

Division needs one more exclusion

For $\frac{f}{g}$ the input must be in both domains **and** must not make g zero. Finding those zeros is usually the real work of the question.

domain of $\frac{f}{g} = (\text{dom } f) \cap (\text{dom } g) \text{ minus } \{x : g(x) = 0\}$

$F(X)$	$G(X)$	$(F/G)(X)$	DOMAIN
$x + 1$	$x - 2$	$\frac{x + 1}{x - 2}$	$x \neq 2$
x^2	$x^2 - 1$	$\frac{x^2}{x^2 - 1}$	$x \neq \pm 1$
$\sqrt{x}$	$x - 3$	$\frac{\sqrt{x}}{x - 3}$	$x \geq 0, x \neq 3$

02 Worked examples

EXAMPLE 1 $f(x) = x + 2$, $g(x) = x^2 - 4$. Find $(f + g)(3)$ and $(f \cdot g)(x)$.

① $f(3) = 5$, $g(3) = 5$

② $(f + g)(3) = 10$

③ $(f \cdot g)(x) = (x + 2)(x^2 - 4)$

④ $= (x + 2)^2(x - 2) = x^3 + 2x^2 - 4x - 8$

ANSWER

10 and $x^3 + 2x^2 - 4x - 8$

EXAMPLE 2 $f(x) = \sqrt{x - 1}$, $g(x) = x - 5$. Find the domain of $\frac{f}{g}$.

① Domain of f : $x \geq 1$

② Domain of g : $\mathbb{R}$

③ Intersection: $x \geq 1$

④ Remove the zero of g : $x = 5$

ANSWER

$x \geq 1$, $x \neq 5$

EXAMPLE 3 $f(x) = 3x$, $g(x) = x - 1$. Solve $(f - g)(x) = 9$.

① $(f - g)(x) = 3x - (x - 1)$
Bracket the whole of g .

② $= 2x + 1$

③ $2x + 1 = 9 \Rightarrow x = 4$

ANSWER

$x = 4$

03 Interactive model

Pick two functions and see the four combinations plotted together.

Combining functions

$f(x) = 2x$, $g(x) = x + 5$. Then $(f + g)(1)$ is:

$(f - g)(x)$ for the same pair is:

Domain of $f + g$ is the ____ of the domains:

$f(x) = 1, g(x) = x - 3$. Domain of $\underline{f} \cdot g$:

$$\mathbb{R}$$

$$x \neq 3$$

$$x > 3$$

$$x \neq 0$$

$f(x) = \sqrt{x}, g(x) = \sqrt{4 - x}$. Domain of $f \cdot g$:

$$x \geq 0$$

$$x \leq 4$$

$$0 \leq x \leq 4$$

$$\mathbb{R}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sum of functions	Funksiyalar yig'indisi	Сумма функций
Difference of functions	Funksiyalar ayirmasi	Разность функций
Product of functions	Funksiyalar ko'paytmasi	Произведение функций
Quotient of functions	Funksiyalar bo'linmasi	Частное функций
Intersection of sets	To'plamlar kesishmasi	Пересечение множеств
Common domain	Umumiy aniqlanish sohasi	Общая область определения
Zero of a function	Funksiya nuli	Ноль функции
Closed under	Yopiq	Замкнуто относительно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$(f \cdot g)(2)$ means:

Dividing functions forbids inputs where:

If both domains are $\mathbb{R}$, the domain of $f - g$ is:

$f(x) = x^2$, $g(x) = -x^2$. Then $f + g$ is:

the zero function

$2x^2$

x^4

undefined

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $f = 3x, g = x + 1$; find $(f + g)(2)$

ANSWER 9

2. Same pair; $(f - g)(2)$

ANSWER 3

3. Same pair; $(f \cdot g)(2)$

ANSWER 18

4. Same pair; $(f/g)(2)$

ANSWER 2

5. $f = x^2, g = 4$; find $(f + g)(x)$

ANSWER $x^2 + 4$

6. Same pair; find $(f \cdot g)(x)$

ANSWER $4x^2$

7. Domain of $f + g$ when both are polynomials

ANSWER $\mathbb{R}$

Medium · the standard question

7 PROBLEMS

1. $f = x + 3, g = x - 3$; find $(f \cdot g)(x)$

ANSWER $x^2 - 9$

2. $f = 2x, g = x^2$; solve $(f + g)(x) = 0$

ANSWER $x = 0$ or $x = -2$

3. $f = \sqrt{x}, g = x$; domain of f/g

ANSWER $x > 0$

4. $f = \frac{1}{x}, g = x - 2$; domain of $f + g$

ANSWER $x \neq 0$

5. $f = x - 1, g = x^2 - 1$; simplify f/g

ANSWER $\frac{1}{x + 1}, x \neq \pm 1$

6. $f = 5, g = x$; find $(f - g)(-3)$

ANSWER 8

7. $f = x^2, g = x$; solve $(f/g)(x) = 4$

ANSWER $x = 4$

Hard · stretch and reason

7 PROBLEMS

1. $f = \sqrt{x-2}, g = \sqrt{8-x}$; domain of $f+g$

ANSWER $2 \leq x \leq 8$

2. Same pair; domain of f/g

ANSWER $2 \leq x < 8$

3. $f = x^2 - 4, g = x + 2$; simplify f/g with its domain

ANSWER $x - 2, x \neq -2$

4. Find f, g with $(f+g)(x) = x^2$ and $(f-g)(x) = 2x$

ANSWER $f = \frac{x^2 + 2x}{2}, g = \frac{x^2 - 2x}{2}$

5. $f = \frac{1}{x-1}, g = \frac{1}{x+1}$; simplify $f-g$

ANSWER $\frac{2}{x^2 - 1}$

6. Show the sum of two even functions is even

ANSWER $(f+g)(-x) = f(-x)+g(-x) = f(x)+g(x)$

7. Is the product of two odd functions odd or even?

ANSWER even

07 Homework

Homework — 5 tasks

State the domain with every combined function, even when it is $\mathbb{R}$.

1. $f(x) = x + 4, g(x) = x^2 - 16$. Find all four combinations.

2. Give the domain of each of those four.

3. $f(x) = \sqrt{x+1}, g(x) = x - 3$. Find the domain of $\frac{f}{g}$.

4. Solve $(f + g)(x) = 0$ for $f(x) = 2x - 1$, $g(x) = x + 7$.
5. Explain why $\frac{x^2 - 1}{x - 1}$ and $x + 1$ are not the same function.

Control work 1, and work on the mistakes

One hour of written assessment on the revision block and the definition of a function; one hour putting right what it exposes.

LESSON 15–16

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 1

P1 · CHAPTER 1–2 REVIEW

LESSON CLOCK

3

37'

10'

25'

5'

Instructions	3 min
The paper	37 min
Self-mark and classify	10 min
Rewrite the wrong ones	25 min
Set the resit list	5 min

LEARNING OBJECTIVES

- Work accurately and independently under time.
- Show the reasoning, not only the answer.
- Classify each mistake as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 48–49
Cambridge	Pure Mathematics 1, pp. 24, 42
Workbook	Control paper A

01

Explanation

The paper — 20 marks, 37 minutes

Q	TASK	MARKS	FROM
1	Solve $x^2 - 7x + 10 = 0$ by factorising	2	L2–3
2	Solve $3x^2 - 5x - 2 = 0$ by formula, exact answers	3	L2–3
3	Write $x^2 - 8x + 21$ in completed-square form and give the vertex	3	L4–5
4	Classify three mappings and say which are functions	3	L8
5	$f(x) = x^2 - 3x$: find $f(-2)$ and $f(a + 1)$	3	L9–10
6	Domain and range of $f(x) = \sqrt{x - 5}$	3	L11–12
7	$f = 2x, g = x - 4$: find $(f/g)(x)$ and its domain	3	L13–14

MARKING

A correct answer with no working scores half. A wrong answer reached by correct method scores most of the marks. Write the method.

Classifying your own mistakes

In the second hour, every learner rules three columns and puts each lost mark in one:

KIND	LOOKS LIKE	WHAT FIXES IT
careless	a sign, a copied digit, an unfinished last line	checking, not revision
method	the right idea applied in the wrong order	one more worked example
knowledge	you did not know the fact or the rule	re-read the section, then five problems

THE POINT OF THE EXERCISE

A learner with six careless marks and a learner with six knowledge marks scored the same and need completely different homework. That is why the column matters more than the total.

The rewrite

Every question that lost a mark is rewritten in full — not corrected in the margin.

Under each rewrite, one sentence: *"I lost this because ..."*.

$$x^2 + 6x + 9 = (x + 3)^2$$

$$\text{so } x^2 + 6x = (x + 3)^2 - 9$$

Question 3 in one picture — the square, the correction, the vertex.

Learners with more than four knowledge marks in one section get that section's problem set as their homework instead of the standard list.

02 Worked examples

EXAMPLE 1 Model answer, Q3: write $x^2 - 8x + 21$ in completed-square form.

1 Half of -8 is -4 .
 $(x - 4)^2 = x^2 - 8x + 16$

2 We need $+21$, we have $+16$.
 Add 5.

3 $x^2 - 8x + 21 = (x - 4)^2 + 5$
 Vertex $(4, 5)$.

ANSWER

$(x - 4)^2 + 5$, vertex $(4, 5)$

EXAMPLE 2 Model answer, Q7: $f = 2x$, $g = x - 4$. Find $(f/g)(x)$ and its domain.

1 $(f/g)(x) = \frac{2x}{x - 4}$

2 Both domains are $\mathbb{R}$.

3 Remove the zero of g : $x = 4$.

ANSWER

$\frac{2x}{x - 4}$, domain $x \neq 4$

EXAMPLE 3 A learner wrote $f(a + 1) = f(a) + 1 = a^2 - 3a + 1$. Name the mistake.

① They added 1 to the output instead of the input.
A method error, not carelessness.

② Correct: $(a + 1)^2 - 3(a + 1)$

③ $= a^2 + 2a + 1 - 3a - 3 = a^2 - a - 2$

ANSWER

Method error; the answer is $a^2 - a - 2$

03 Interactive model

Re-do the two questions the class lost most marks on before the rewrite starts.

Mark these six answers

$x^2 - 7x + 10 = (x - 2)(x - 5)$ — is it right?

yes

no, sign error

no, wrong numbers

no, it does not factorise

For $3x^2 - 5x - 2$, $D = 25 - 24 = 1$. Right?

yes

no — $D = 49$

no — $D = 1$ but sign wrong

no — $D = -49$

$x^2 - 8x + 21 = (x - 4)^2 + 21$

right

careless

method error — forgot to subtract 16

knowledge gap

Domain of $\sqrt{x-5}$ given as $x > 5$

right

careless — should be $x \geq 5$

method error

right, both work

$(f/g)(x) = \frac{2x}{x-4}$ with domain $\mathbb{R}$

right

careless

method — forgot $x \neq 4$

knowledge gap

$x \mapsto \pm\sqrt{x}$ called a function

right

knowledge gap

careless

method error

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Assessment criteria	Baholash mezonlari	Критерии оценивания
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo'shlig'i	Пробел в знаниях
Working (shown)	Yechim yozuvi	Ход решения
Mark scheme	Baholash sxemasi	Схема оценивания
Correction	Tuzatish	Исправление
Resit task	Qayta topshiriq	Повторное задание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A correct answer with no working scores:

full marks

half

nothing

one mark

A dropped minus sign is best classed as:

careless

method

knowledge

unavoidable

Not knowing what a discriminant is, is:

careless

method

knowledge

careless

The most useful number from a control work is:

- the total
- the class average
- the column your losses fell in
- the rank

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $x^2 - 7x + 10 = 0$

ANSWER $x = 2, 5$

2. Solve $x^2 - 9 = 0$

ANSWER $x = \pm 3$

3. Complete the square: $x^2 + 2x$

ANSWER $(x + 1)^2 - 1$

4. $f(x) = x^2 - 3x$; find $f(-2)$

ANSWER 10

5. Domain of $\sqrt{x - 5}$

ANSWER $x \geq 5$

6. Is $x \mapsto x^2$ a function?

ANSWER yes

7. $f = 2x$, $g = x - 4$; find $(f + g)(5)$

ANSWER 11

Medium · the standard question

7 PROBLEMS

1. Solve $3x^2 - 5x - 2 = 0$

ANSWER $x = 2, -\frac{1}{3}$

2. Complete the square: $x^2 - 8x + 21$

ANSWER $(x - 4)^2 + 5$

3. $f(x) = x^2 - 3x$; find $f(a + 1)$

ANSWER $a^2 - a - 2$

4. Range of $y = (x - 4)^2 + 5$

ANSWER $y \geq 5$

5. Domain of $\frac{2x}{x - 4}$

ANSWER $x \neq 4$

6. Classify $x \mapsto x^2 + 1$ on $\mathbb{R}$

ANSWER many-to-one

7. Solve $2x^2 + 7x + 3 = 0$

ANSWER $x = -3, -0.5$

Hard · stretch and reason

7 PROBLEMS

1. For what k does $x^2 + kx + 16 = 0$ have one root?

ANSWER $k = \pm 8$

2. Range of $y = x^2 - 8x + 21$ on $2 \leq x \leq 5$

ANSWER $5 \leq y \leq 9$

3. $f(x) = x^2$; simplify $\frac{f(x+h) - f(x)}{h}$

ANSWER $2x + h$

4. Domain of $\frac{\sqrt{x-1}}{x-3}$

ANSWER $x \geq 1, x \neq 3$

5. Solve $x^4 - 5x^2 + 4 = 0$

ANSWER $x = \pm 1, \pm 2$

6. Find a, b with $x^2 + ax + b = (x-3)^2 - 4$

ANSWER $a = -6, b = 5$

7. Explain why $D < 0$ means no x -intercepts

ANSWER The parabola never reaches the axis

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. It is not optional.

1. Rewrite in full every question that lost a mark, with the “I lost this because ...” sentence.
2. Five problems from the section your knowledge column was heaviest in.
3. Solve $2x^2 - 9x + 4 = 0$ twice — factorising and by formula — and check they agree.

4. Find the domain and range of $f(x) = \frac{3}{x+1}$.

Composite, inverse and periodic functions

Three ways one function can be built from another — chaining it, undoing it, and repeating it.

LESSON 17–18

2 × 40 MIN

ALGEBRA 10, §1.5

P1 · 2.4–2.6

LESSON CLOCK

10'

12'

16'

14'

8'

5'

Composition — inner first	10 min
fg is not gf	12 min
Inverses	16 min
Periodic functions	14 min
Practice	8 min
Homework	5 min

LEARNING OBJECTIVES

- Form and evaluate the composite functions fg and gf .
- Show that $fg \neq gf$ in general.
- Find the inverse of a one-to-one function and state its domain and range.
- Recognise a periodic function and state its period.

TEXTBOOK

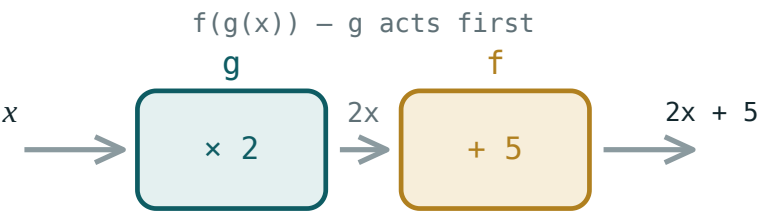
National	pp. 50–58
Cambridge	Pure Mathematics 1, pp. 42–55
Workbook	P1 Exercise 2D, 2E

01

Explanation

Composition — the inner function goes first

$fg(x)$ means $f(g(x))$: apply g , then feed the result into f . The function written **closest to the x** acts first.



Two machines in a line. The output of g becomes the input of f .

$f(x) = x^2, g(x) = x + 3 \Rightarrow fg(x) = (x + 3)^2 \text{ but } gf(x) = x^2 + 3$

ORDER MATTERS

fg and gf are different functions. They agree only for special pairs — and finding the x where they happen to be equal is a standard exam question.

The domain of fg is the set of x in the domain of g whose image $g(x)$ lies in the domain of f .

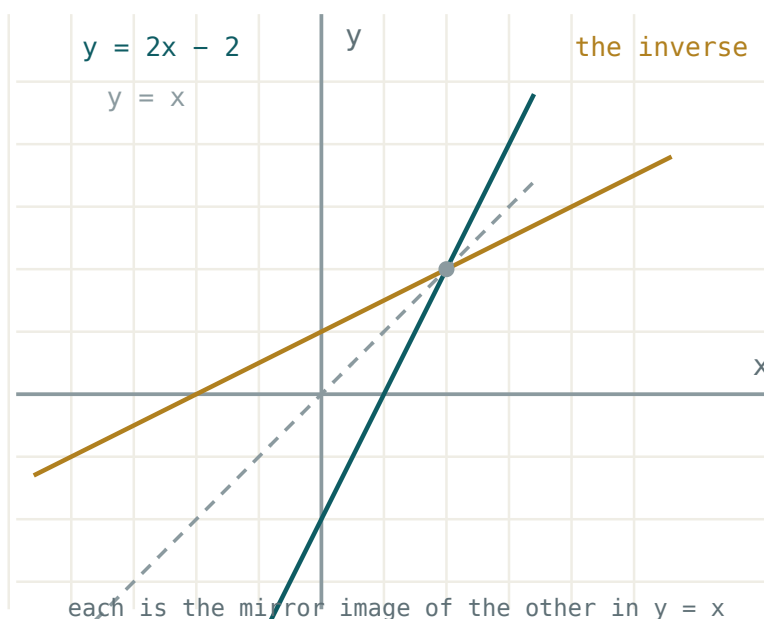
Inverses — undoing the machine

ONLY ONE-TO-ONE FUNCTIONS HAVE INVERSES

If two inputs share an output, the reverse arrow does not know where to go. That is why $f(x) = x^2$ on $\mathbb{R}$ has no inverse, but on $x \geq 0$ it has one.

The method is always the same three steps:

1. Write $y = f(x)$.
2. Make x the subject.
3. Swap the letters — and swap domain with range.



*The inverse is the mirror image in the line $y = x$.
Domain and range swap places.*

$$f(x) = 3x - 5 \Rightarrow y = 3x - 5 \Rightarrow x = \frac{y + 5}{3} \Rightarrow f^{-1}(x) = \frac{x + 5}{3}$$

f^{-1} IS NOT $\frac{1}{f}$

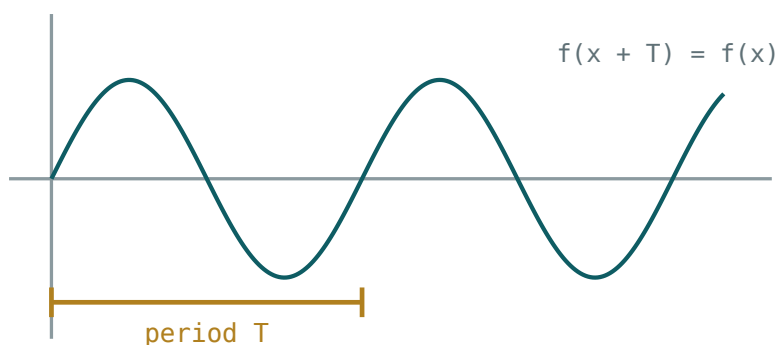
The -1 is notation for “inverse”, not an index. $f^{-1}(x)$ and $[f(x)]^{-1}$ are different things.

The defining property: $f^{-1}f(x) = x$ and $ff^{-1}(x) = x$. Composing a function with its inverse gives the identity — that is the check to run on every answer.

Periodic functions

A function is **periodic** when its whole graph repeats after a fixed horizontal shift:

$f(x + T) = f(x)$ for every x , with the smallest such $T > 0$ called the period



One block of the graph, repeated forever. The period is the width of the block.

Trigonometric functions are the standard examples — $\sin x$ and $\cos x$ have period 360° , $\tan x$ has period 180° . So do many real quantities: hours of daylight, tides, the current in an alternating circuit.

SMALLEST

$\sin(x + 720^\circ) = \sin x$ too — but 720° is not the period, because 360° works and is smaller.

02 Worked examples

EXAMPLE 1 $f(x) = 2x + 1$, $g(x) = x^2$. Find $fg(3)$ and $gf(3)$.

① $g(3) = 9$

Inner first.

② $fg(3) = f(9) = 19$

③ $f(3) = 7$

④ $gf(3) = g(7) = 49$

Different, as expected.

ANSWER

$$fg(3) = 19, gf(3) = 49$$

EXAMPLE 2 Find f^{-1} for $f(x) = \frac{x+2}{x-1}$, $x \neq 1$.

① $y = \frac{x+2}{x-1}$

② $y(x-1) = x+2 \Rightarrow yx - y = x+2$

③ $x(y-1) = y+2 \Rightarrow x = \frac{y+2}{y-1}$

④ Swap letters.

This function is its own inverse.

ANSWER

$$f^{-1}(x) = \frac{x+2}{x-1}, x \neq 1$$

EXAMPLE 3 $f(x) = x + 4$, $g(x) = 3x$. Solve $fg(x) = gf(x)$.

① $fg(x) = 3x + 4$

② $gf(x) = 3(x + 4) = 3x + 12$

③ $3x + 4 = 3x + 12 \Rightarrow 4 = 12$

A contradiction.

ANSWER

No solution — they are never equal

03 Interactive model

Feed a value through both orders and watch the two answers separate.

Inner first, outer second

$f(x) = x + 1$, $g(x) = 2x$. Then $fg(3)$ is:

Same pair; $gf(3)$ is:

The inverse of $f(x) = 5x$ is:

$f(x) = x^2$ on $\mathbb{R}$ has an inverse because:

it does not

it is one-to-one

it is continuous

it is even

The graph of f^{-1} is the graph of f reflected in:

the x-axis

the y-axis

$y = x$

the origin

If $f(x + 6) = f(x)$ for all x , the period could be:

6

12

3

any of these

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Composite function	Murakkab funksiya	Сложная функция
Inner function	Ichki funksiya	Внутренняя функция
Outer function	Tashqi funksiya	Внешняя функция
Inverse function	Teskari funksiya	Обратная функция
Identity function	Ayniy funksiya	Тождественная функция
Periodic function	Davriy funksiya	Периодическая функция
Period	Davr	Период
One-to-one	Biror-birga	Взаимно однозначное
Reflection in $y = x$	$y = x$ ga nisbatan simmetriya	Симметрия относительно $y = x$

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$fg(x)$ means:

$ff^{-1}(x)$ equals:

A function has an inverse when it is:

The period of $\cos x$ is:

90°

180°

360°

720°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $f = x + 2, g = 3x$; find $fg(1)$

ANSWER 5

2. Same pair; $gf(1)$

ANSWER 9

3. Inverse of $f(x) = x + 7$

ANSWER $x - 7$

4. Inverse of $f(x) = 4x$

ANSWER $\frac{x}{4}$

5. Period of $\sin x$

ANSWER 360°

6. Period of $\tan x$

ANSWER 180°

7. $f^{-1}f(5)$ equals

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. $f = x^2, g = x - 1$; find $fg(x)$

ANSWER $(x - 1)^2$

2. Same pair; $gf(x)$

ANSWER $x^2 - 1$

3. Inverse of $f(x) = 2x - 9$

ANSWER $\frac{x + 9}{2}$

4. Inverse of $f(x) = \frac{1}{x}$

ANSWER $\frac{1}{x}$

5. $f = 3x, g = x + 4$; solve $fg(x) = 0$

ANSWER $x = -4$

6. Inverse of $f(x) = x^3 + 1$

ANSWER $\sqrt[3]{x - 1}$

7. If $f(x + 5) = f(x)$ and no smaller shift works, the period is

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. $f = \sqrt{x}, g = x - 4$; domain of fg

ANSWER $x \geq 4$

2. Same pair; domain of gf

ANSWER $x \geq 0$

3. Inverse of $f(x) = \frac{2x + 1}{x - 3}$

ANSWER $\frac{3x + 1}{x - 2}$

4. $f = x + 3, g = x^2$; solve $fg(x) = gf(x)$

ANSWER $x = -1$

5. Restrict $f(x) = x^2 - 4x$ so that it has an inverse

ANSWER $x \geq 2$ (or $x \leq 2$)

6. Find f^{-1} for that restriction

ANSWER $2 + \sqrt{x + 4}$

7. Show that if f has period T then so does $f(x) + 5$

ANSWER Adding a constant does not change the repeat

07 Homework

Homework — 5 tasks

Check every inverse by composing it with the original.

- $f(x) = 3x - 2, g(x) = x^2$. Find $fg(x)$ and $gf(x)$.
- Find f^{-1} for $f(x) = 5 - 2x$ and check $f^{-1}f(4) = 4$.
- Find f^{-1} for $f(x) = \frac{x}{x + 1}, x \neq -1$.
- Explain why $f(x) = x^4$ has no inverse on $\mathbb{R}$, and give a domain where it does.

5. Sketch a periodic function with period 4 that is not trigonometric.

Properties of a function

Zeros, sign, monotonicity, parity and boundedness — the five questions asked of every function for the rest of the course.

LESSON 19–21

3 × 40 MIN

ALGEBRA 10, §1.6

P1 · 2.2, 3.1

LESSON CLOCK

12

20'

20'

14'

55'

14'

Zeros and the sign chart	12 min
Increasing and decreasing	20 min
Even and odd	20 min
Boundedness	14 min
Practice across all five	55 min
Homework	14 min

LEARNING OBJECTIVES

- Find the zeros of a function and the intervals where it is positive or negative.
- Decide whether a function is increasing or decreasing on an interval.
- Test a function for even and odd symmetry algebraically.
- Say whether a function is bounded above, below, or neither.

TEXTBOOK

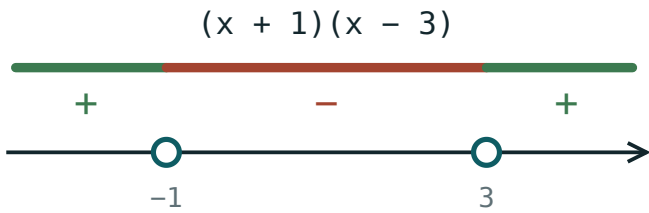
National	pp. 59–70
Cambridge	Pure Mathematics 1, pp. 30–36, 58–62
Workbook	P1 Exercise 2B, 3A

01

Explanation

Zeros and the sign chart

A **zero** of f is an input with $f(x) = 0$ — where the graph meets the x -axis. Between consecutive zeros a continuous function cannot change sign, so the zeros cut the domain into intervals on which the sign is constant.



negative between the roots, positive outside

The zeros split the line into three intervals. Test one point in each and the whole interval follows.

THE METHOD, IN FOUR STEPS

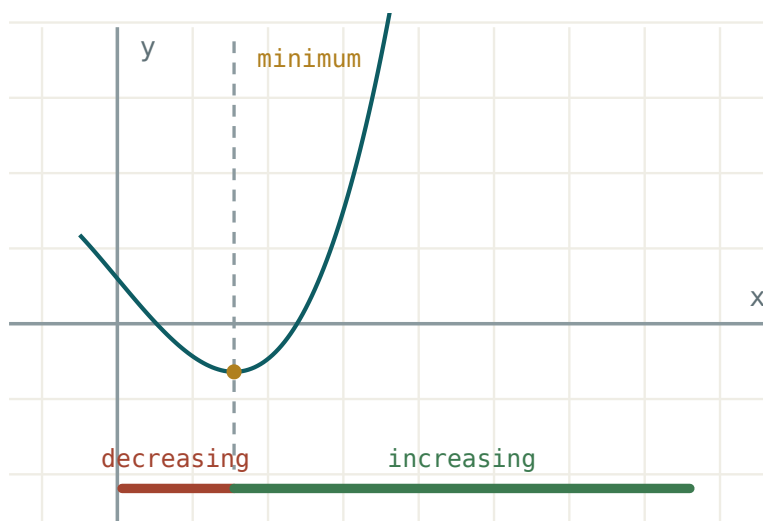
Find the zeros · mark them on a number line, together with any excluded points · test one convenient point in each interval · write the answer as intervals.

For $f(x) = x^2 - x - 6 = (x - 3)(x + 2)$ the zeros are -2 and 3 . Testing $x = 0$ gives -6 , so $f < 0$ on $(-2, 3)$ and $f > 0$ outside it.

Increasing and decreasing

THE DEFINITIONS

f is **increasing** on an interval when $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$ for every pair in it, and **decreasing** when $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$. A function that is one or the other on the whole domain is **monotonic**.



Decreasing to the left of the vertex, increasing to the right. Neither word describes the whole domain.

ALWAYS NAME THE INTERVAL

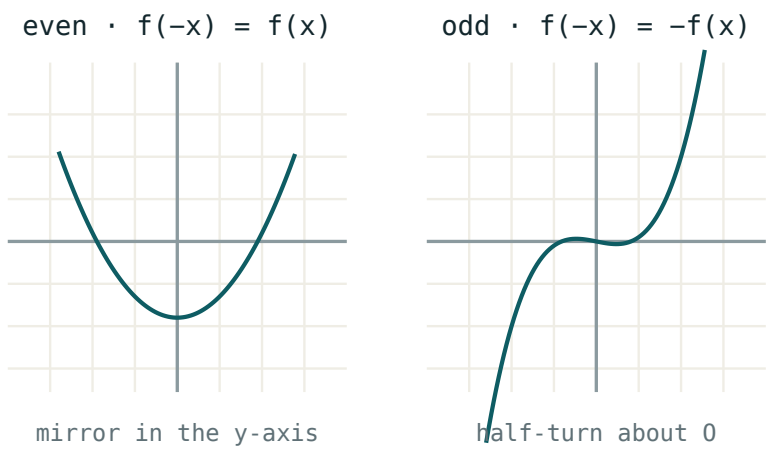
" $y = x^2$ is increasing" is false. " $y = x^2$ is increasing on $x \geq 0$ " is true. Monotonicity is a property of a function **on an interval**, never of a function alone.

A monotonic function is automatically one-to-one, and therefore has an inverse — which is why this property matters so much in the next chapter.

Even and odd

Two symmetries are worth naming, because they halve the work of sketching:

$$\text{even: } f(-x) = f(x) \quad \text{odd: } f(-x) = -f(x)$$



An even graph folds onto itself in the y-axis. An odd graph maps onto itself under a half-turn about the origin.

FUNCTION	$F(-X)$	VERDICT
x^2	x^2	even
x^3	$-x^3$	odd
$x^2 + x$	$x^2 - x$	neither
$ x $	$ x $	even
$\frac{1}{x}$	$-\frac{1}{x}$	odd

"NEITHER" IS A REAL ANSWER

Most functions are neither even nor odd. The only function that is both is $f(x) = 0$. Also: the domain must itself be symmetric about 0, or the question cannot even be asked.

Boundedness

f is **bounded above** if there is a number M with $f(x) \leq M$ for every x in the domain, and **bounded below** if some m has $f(x) \geq m$. Bounded on both sides is simply **bounded**.

FUNCTION	BELOW	ABOVE
x^2	yes, by 0	no
$-x^2 + 5$	no	yes, by 5
$\sin x$	yes, by -1	yes, by 1
$2x + 1$	no	no
$\frac{1}{1 + x^2}$	yes, by 0	yes, by 1

Boundedness is exactly a statement about the **range**: bounded above means the range has a ceiling. So finding the range answers this property for free.

02

Worked examples

EXAMPLE 1

Find the zeros and the sign intervals of $f(x) = x^3 - 4x$.

1

$x(x^2 - 4) = x(x - 2)(x + 2)$
Factorise fully.

2

Zeros $-2, 0, 2$.
Four intervals.

3

Test $-3, -1, 1, 3$: $-15, 3, -3, 15$.

4

Signs alternate.

ANSWER

$f > 0$ on $(-2, 0) \cup (2, \infty)$; $f < 0$ on $(-\infty, -2) \cup (0, 2)$

EXAMPLE 2 Decide whether $f(x) = x^3 - 3x$ is even, odd or neither.

① $f(-x) = (-x)^3 - 3(-x)$
Substitute, do not guess.

② $= -x^3 + 3x$

③ $= -(x^3 - 3x) = -f(x)$

ANSWER

Odd

EXAMPLE 3 On which intervals is $f(x) = x^2 - 6x + 5$ decreasing, and is it bounded?

① Complete the square: $(x - 3)^2 - 4$.
Vertex $(3, -4)$.

② A positive parabola falls to the left of its vertex.
Decreasing on $x \leq 3$.

③ Range $y \geq -4$.
Bounded below only.

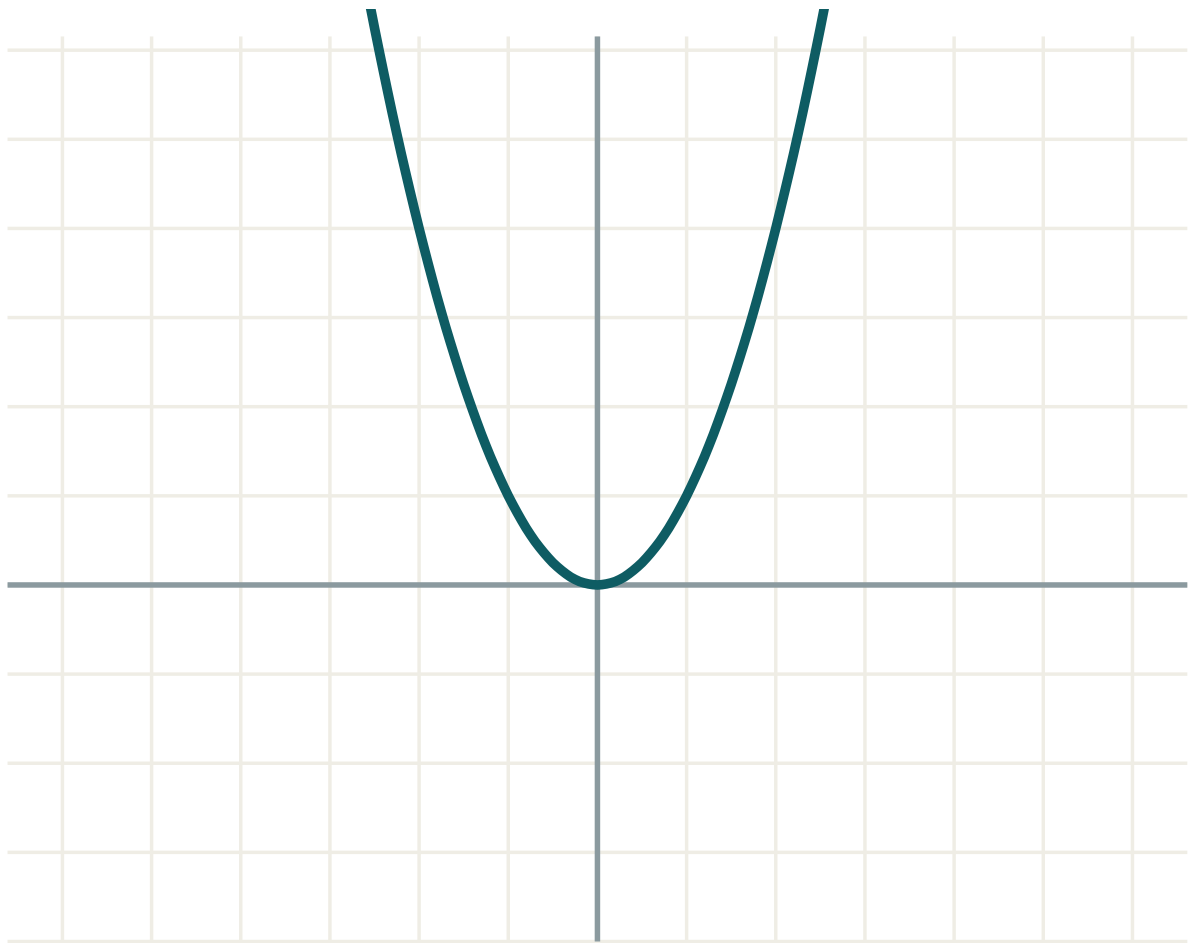
ANSWER

Decreasing on $(-\infty, 3]$; bounded below by -4 , not above

03 Interactive model

Move the point along the curve and watch the sign, the direction and the symmetry all reported at once.

Five properties on one curve



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Zero of a function	Funksiya nuli	Ноль функции
Sign of a function	Funksiya ishorasi	Знак функции
Increasing function	O'suvchi funksiya	Возрастающая функция
Decreasing function	Kamayuvchi funksiya	Убывающая функция
Monotonic	Monoton	Монотонная
Even function	Juft funksiya	Чётная функция
Odd function	Toq funksiya	Нечётная функция
Bounded above	Yuqoridan chegaralangan	Ограничена сверху
Bounded below	Quyidan chegaralangan	Ограничена снизу
Interval of constant sign	Ishora saqlanish oralig'i	Промежуток знакопостоянства

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$f(-x) = f(x)$ means the function is:

$y = x^2$ is increasing on:

The zeros of $f(x) = (x - 1)(x + 4)$ are:

$f(x) = \sin x$ is:

bounded

bounded below only

bounded above only

unbounded

A monotonic function is always:

even

periodic

one-to-one

bounded

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Zeros of $f(x) = x - 5$

ANSWER $x = 5$

2. Zeros of $f(x) = x^2 - 16$

ANSWER $x = \pm 4$

3. Is x^4 even or odd?

ANSWER even

4. Is x^5 even or odd?

ANSWER odd

5. Is $y = 3x + 1$ increasing or decreasing?

ANSWER increasing

6. Is $y = 7 - 2x$ increasing or decreasing?

ANSWER decreasing

7. Is $y = x^2$ bounded below?

ANSWER yes, by 0

Medium · the standard question

7 PROBLEMS

1. Sign intervals of $(x - 1)(x - 4)$

ANSWER > 0 outside $[1, 4]$, < 0 inside

2. Is $f(x) = x^2 + x^4$ even, odd or neither?

ANSWER even

3. Is $f(x) = x^3 + x$ even, odd or neither?

ANSWER odd

4. Is $f(x) = x^2 + x$ even, odd or neither?

ANSWER neither

5. Where is $y = (x + 2)^2 - 1$ increasing?

ANSWER $x \geq -2$

6. Range and boundedness of $y = 4 - x^2$

ANSWER $y \leq 4$, bounded above only

7. Zeros of $f(x) = x^3 - x$

ANSWER $x = 0, \pm 1$

Hard · stretch and reason

7 PROBLEMS

1. Sign intervals of $\frac{x-1}{x+2}$

ANSWER > 0 on $(-\infty, -2) \cup (1, \infty)$

2. Show $f(x) = \frac{1}{x}$ is odd

ANSWER $f(-x) = -\frac{1}{x} = -f(x)$

3. Is $f(x) = \sqrt{x}$ even, odd or neither? Why?

ANSWER Neither — the domain is not symmetric

4. Prove the sum of two odd functions is odd

ANSWER $(f+g)(-x) = -f(x) - g(x) = -(f+g)(x)$

5. Prove the product of an even and an odd function is odd

ANSWER $f(-x)g(-x) = f(x) \cdot (-g(x))$

6. Find all k making $y = x^2 + kx$ increasing on $x \geq 0$

ANSWER $k \geq 0$

7. Show $\frac{1}{1+x^2}$ is bounded, and give both bounds

ANSWER $0 < y \leq 1$

07 Homework

Homework — 6 tasks

Every parity answer must show $f(-x)$ worked out, not just the verdict.

- Zeros and sign intervals of $f(x) = x^2 - 3x - 10$.
- Test $f(x) = x^4 - 3x^2$ and $g(x) = x^5 - x$ for parity.
- State the intervals on which $y = -x^2 + 8x$ increases and decreases.

4. Say whether each of x^3 , $|x|$ and $\frac{1}{x^2}$ is bounded, and how.
5. Give a function that is increasing on $\mathbb{R}$ and odd.
6. Explain why an even function can never be one-to-one on a symmetric domain containing more than one point.

Transformations of the graph of a function

Four moves — shift, shift, stretch, reflect — and the rule that inside the bracket everything works backwards.

LESSON 22–23

2 × 40 MIN

ALGEBRA 10, §1.7

P1 · 3.1–3.4

LESSON CLOCK

10'

16'

16'

16'

22'

10'

Outside the bracket — what you expect	10 min
Inside the bracket — backwards	16 min
Stretches	16 min
Reflections and combinations	16 min
Practice	22 min
Homework	10 min

LEARNING OBJECTIVES

- Sketch $y = f(x) + a$ and $y = f(x - a)$ from the graph of f .
- Sketch $y = a \cdot f(x)$ and $y = f(ax)$.
- Sketch $y = -f(x)$ and $y = f(-x)$.
- Combine two transformations and give them in the correct order.

TEXTBOOK

National	pp. 71–80
Cambridge	Pure Mathematics 1, pp. 58–74
Workbook	P1 Exercise 3A–3C

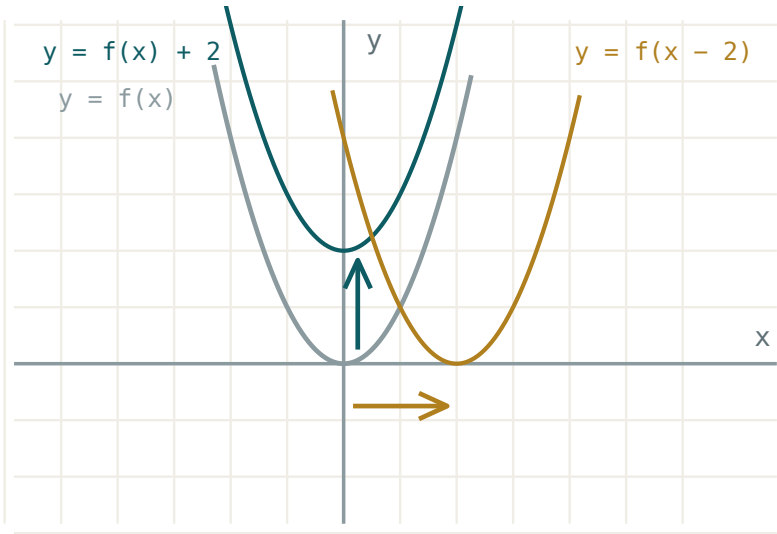
01

Explanation

Outside the bracket behaves

Everything done to $f(x)$ **after** the function has acted moves the graph vertically, in the direction you would guess:

EQUATION	EFFECT	A POINT (p, q) GOES TO
$y = f(x) + a$	up by a	$(p, q + a)$
$y = a \cdot f(x)$	stretch, factor a , from the x -axis	(p, aq)
$y = -f(x)$	reflect in the x -axis	$(p, -q)$



The same curve, shifted. Only the y-coordinates changed.

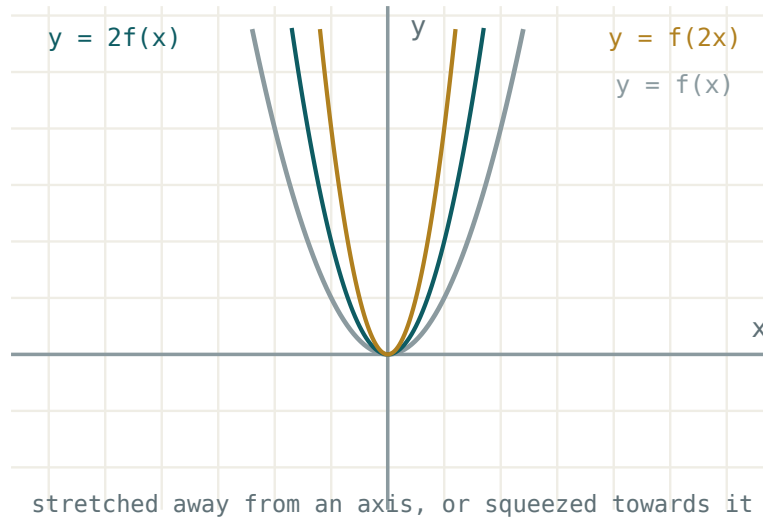
Inside the bracket works backwards

THE ONE RULE TO MEMORISE

Anything done to x **before** the function acts moves the graph horizontally and does the **opposite** of what it looks like. $f(x - 3)$ moves the graph **right** 3, not left.

Why: the new graph at $x = 3$ shows the old value at 0. To see the old picture you must travel further along, so the whole graph slides right.

EQUATION	EFFECT	A POINT (p, q) GOES TO
$y = f(x - a)$	right by a	$(p + a, q)$
$y = f(ax)$	compress, factor $\frac{1}{a}$, towards the y-axis	$(\frac{p}{a}, q)$
$y = f(-x)$	reflect in the y-axis	$(-p, q)$



A stretch of factor 2 from the x-axis, and a compression of factor $\frac{1}{2}$ towards the y-axis. Different curves.

$f(2x)$ IS NOT A STRETCH BY 2

It is a compression: every x -coordinate is **halved**. $f\left(\frac{x}{2}\right)$ is the one that stretches by 2.

Combining, and the order

$y = 3f(x - 2) + 1$ is three transformations. Outside-the-bracket operations are applied in ordinary BIDMAS order — stretch first, then shift:

1. $f(x - 2)$: right 2.
2. $3f(x - 2)$: stretch factor 3 from the x -axis.
3. $3f(x - 2) + 1$: up 1.

ORDER CHANGES THE ANSWER

Stretch-then-shift and shift-then-stretch give different curves. $3f(x) + 1$ and $3[f(x) + 1]$ are not the same: the second is $3f(x) + 3$.

Completed-square form is exactly this idea for quadratics: $y = (x - 3)^2 + 2$ says “take $y = x^2$, move it right 3 and up 2” — which is why the vertex is $(3, 2)$.

02 Worked examples

EXAMPLE 1 The point $(2, 5)$ is on $y = f(x)$. Where is it on $y = 2f(x + 1) - 3$?

- ① $x + 1$ inside: move left 1.
 x -coordinate $2 - 1 = 1$.

- ② Stretch by 2: $5 \cdot 2 = 10$.

- ③ Down 3: $10 - 3 = 7$.

ANSWER

$(1, 7)$

EXAMPLE 2 Describe the transformation taking $y = x^2$ to $y = (x + 4)^2 - 9$.

- ① $(x + 4)$ inside: left 4.
 Backwards, as always.

- ② $- 9$ outside: down 9.

- ③ Vertex moves from $(0, 0)$ to $(-4, -9)$.

ANSWER

Translation left 4 and down 9

EXAMPLE 3 Write $y = x^2 - 10x + 18$ as a transformation of $y = x^2$.

- ① Complete the square.

$$(x - 5)^2 - 25 + 18$$

- ② $= (x - 5)^2 - 7$

- ③ Read off the two moves.

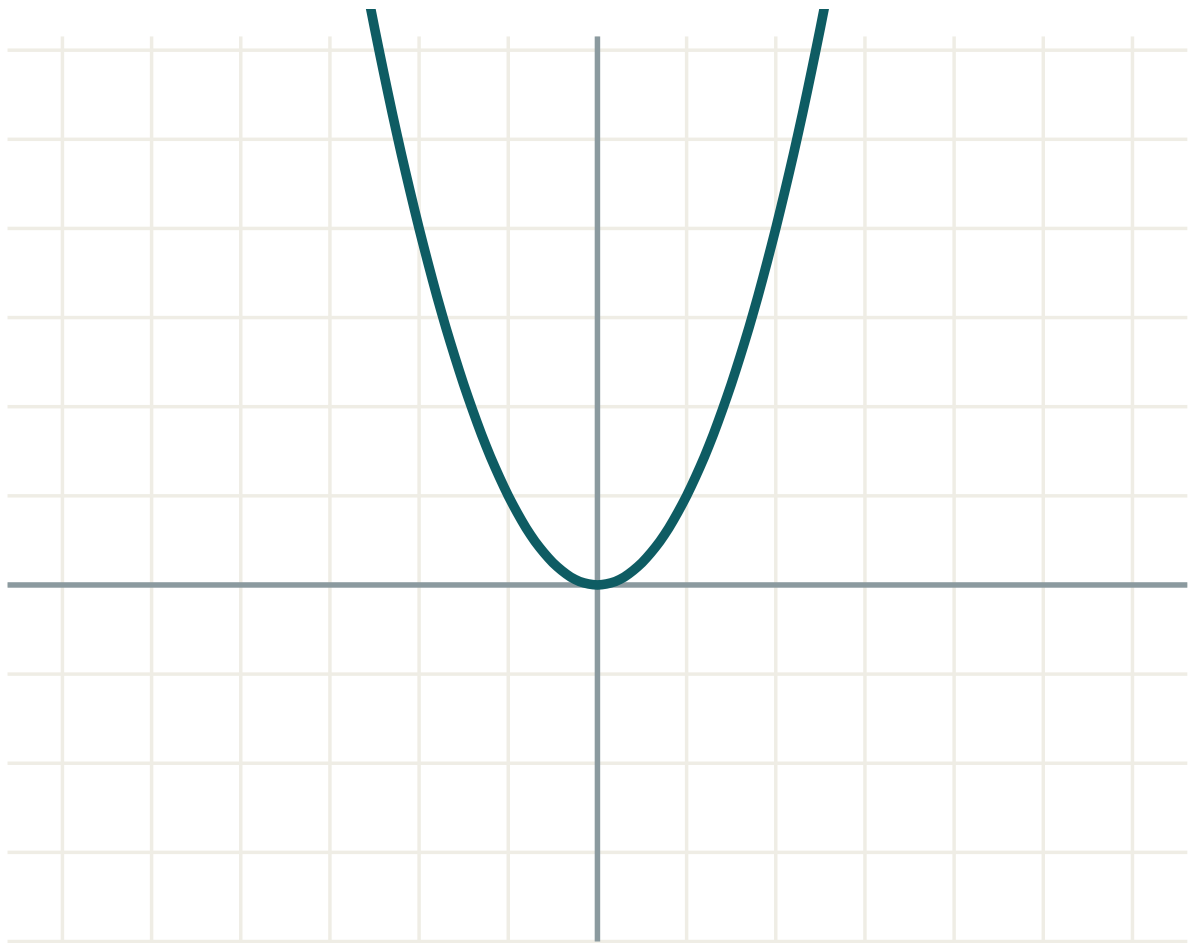
ANSWER

Right 5, down 7 — vertex $(5, -7)$

03 Interactive model

Drag a, b and c and watch which coordinates move.

$$y = a \cdot f(x - b) + c$$



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Transformation	Almashtirish	Преобразование
Translation	Parallel ko'chirish	Параллельный перенос
Vertical shift	Vertikal siljish	Вертикальный сдвиг
Horizontal shift	Gorizontal siljish	Горизонтальный сдвиг
Stretch	Cho'zish	Растяжение
Compression	Siqish	Сжатие
Scale factor	Cho'zish koeffitsienti	Коэффициент растяжения
Reflection	Simmetriya	Отражение
Invariant point	O'zgarmas nuqta	Неподвижная точка
Image of a point	Nuqta tasviri	Образ точки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$y = f(x) + 5$ moves the graph:

up 5

down 5

left 5

right 5

$y = f(x - 5)$ moves the graph:

left 5

right 5

up 5

down 5

$y = f(3x)$ is a:

stretch factor 3

compression factor $\frac{1}{3}$

shift right 3

reflection

$y = -f(x)$ reflects in:

the y -axis

the x -axis

$y = x$

the origin

The vertex of $y = (x + 1)^2 + 6$ is:

$(1, 6)$

$(-1, 6)$

$(-1, -6)$

$(1, -6)$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Describe $y = f(x) + 3$

ANSWER up 3

2. Describe $y = f(x) - 7$

ANSWER down 7

3. Describe $y = f(x - 2)$

ANSWER right 2

4. Describe $y = f(x + 6)$

ANSWER left 6

5. Describe $y = 4f(x)$

ANSWER stretch factor 4 from the x -axis

6. Describe $y = -f(x)$

ANSWER reflect in the x -axis

7. Vertex of $y = (x - 1)^2$

ANSWER $(1, 0)$

Medium · the standard question

7 PROBLEMS

1. Vertex of $y = (x + 3)^2 - 5$

ANSWER $(-3, -5)$

2. $(4, 2)$ on $y = f(x)$; find it on $y = f(x) + 6$

ANSWER $(4, 8)$

3. Same point on $y = f(x - 1)$

ANSWER $(5, 2)$

4. Same point on $y = 3f(x)$

ANSWER $(4, 6)$

5. Same point on $y = f(2x)$

ANSWER $(2, 2)$

6. Write $y = x^2 + 6x + 4$ as a transformation of $y = x^2$

ANSWER left 3, down 5

7. Describe $y = f(-x)$

ANSWER reflect in the y -axis

Hard · stretch and reason

7 PROBLEMS

1. $(3, -1)$ on $y = f(x)$; find it on $y = 2f(x + 3) - 4$

ANSWER $(0, -6)$

2. Describe $y = -2f(x) + 1$ in order

ANSWER stretch 2, reflect in x -axis, up 1

3. Write $y = 2x^2 - 12x + 20$ in the form $a(x - h)^2 + k$

ANSWER $2(x - 3)^2 + 2$

4. What single transformation maps $y = x^2$ to $y = 4x^2$?

ANSWER stretch factor 4 from the x -axis (or compression $\frac{1}{2}$ towards the y -axis)

5. Find a, b with $f(x) = (x - a)^2 + b$ passing through $(0, 10)$ and vertex on $x = 3$

ANSWER $a = 3, b = 1$

6. Which points of $y = f(x)$ are unmoved by $y = -f(x)$?

ANSWER those with $y = 0$

7. Which points are unmoved by $y = f(-x)$?

ANSWER those with $x = 0$

07 Homework

Homework — 5 tasks

Sketch, do not only describe. One axis set per question is enough.

- Sketch $y = x^2$, $y = (x - 2)^2$ and $y = (x - 2)^2 + 3$ on one set of axes.
- Describe the transformation from $y = x^2$ to $y = x^2 - 4x + 1$.
- $(-2, 7)$ is on $y = f(x)$. Find its image on $y = 3f(x + 2) - 1$.
- Explain in your own words why $f(x - 3)$ moves the graph right.

5. Sketch $y = -f(x)$ and $y = f(-x)$ for a curve of your choice and say how they differ.

Modelling with linear and quadratic functions

Turning a paragraph of Uzbek life into a function, solving it, and then checking the answer means something.

LESSON 24–25

2 × 40 MIN

ALGEBRA 10, §1.8

P1 · 1.7, 3.5

LESSON CLOCK

8'

16'

22'

16'

20'

8'

The four-step cycle	8 min
Linear models	16 min
Quadratic models and the vertex	22 min
Judging a model	16 min
Practice	20 min
Homework	8 min

LEARNING OBJECTIVES

- Define variables clearly and write a function from a described situation.
- State the domain that the context — not the algebra — allows.
- Solve the model and interpret the solution in the original words.
- Judge whether a model is reasonable and name one of its limitations.

TEXTBOOK

National	pp. 81–88
Cambridge	Pure Mathematics 1, pp. 18–23, 75–79
Workbook	P1 Exercise 1F, 3D

01

Explanation

The four-step cycle

NEVER SKIP STEP 1

1 Define — “let x be the width in metres”. **2 Translate** — build the function and state the domain the situation allows. **3 Solve** — ordinary algebra. **4 Interpret** — answer in the words of the question, with units.

Most lost marks in modelling questions are lost at step 1 and step 4, not step 3. A number with no units is not an answer, and “ $x = 4$ ” does not tell a reader what is 4 metres long.

Linear models — a fixed part and a rate

Any situation with a starting amount and a constant rate of change is linear:

$$y = (\text{starting amount}) + (\text{rate}) \times (\text{number of units})$$

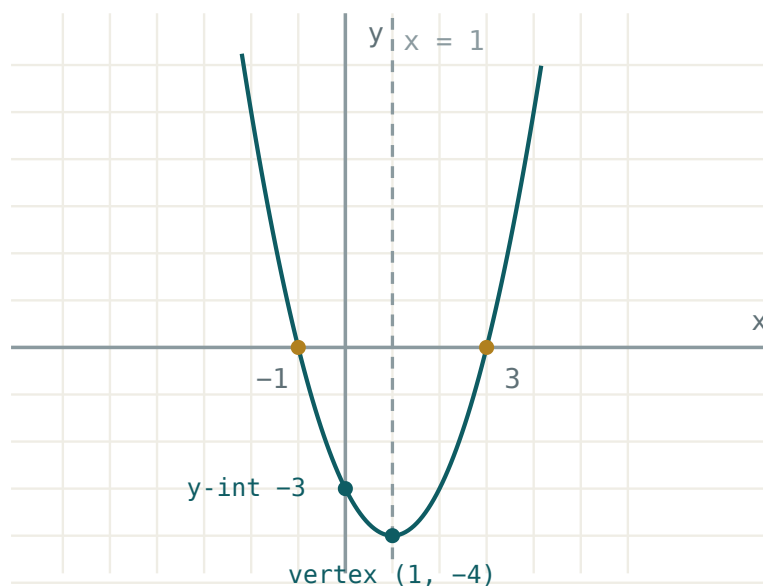
SITUATION	MODEL	WHAT C AND M MEAN
taxi: 8 000 so'm plus 2 500 a km	$C = 8000 + 2500d$	flag fall; price per km
a 40 000 so'm card losing 300 a minute	$B = 40000 - 300t$	credit; cost per minute
a tank filling at 12 litres a minute from 50 litres	$V = 50 + 12t$	start volume; flow rate

THE DOMAIN COMES FROM THE WORLD

$B = 40000 - 300t$ is algebraically fine at $t = 500$, but the card empties at $t = 133\frac{1}{3}$. The domain is $0 \leq t \leq 133.3$, and saying so is worth a mark.

Quadratic models — where the vertex is the answer

A quadratic appears whenever two quantities that vary must be multiplied — an area from two lengths, a revenue from price times quantity, a height under gravity.



For a negative parabola the vertex is the maximum. Its x tells you what to choose; its y tells you what you get.

Fencing. 60 m of fence, a rectangular pen against a wall (three sides fenced). Let the two equal sides be x . Then the third is $60 - 2x$ and

$$A(x) = x(60 - 2x) = 60x - 2x^2 = -2(x - 15)^2 + 450$$

so the greatest area is 450 m^2 when $x = 15$ — and the domain is $0 < x < 30$, because outside it a side would be negative.

WHICH COORDINATE ANSWERS THE QUESTION?

“What width gives the largest area?” wants the x of the vertex. “What is the largest area?” wants the y . Read the question twice before writing the number.

Judging the model

Every model makes assumptions. Naming one is part of a complete answer:

MODEL	AN ASSUMPTION	WHERE IT FAILS
taxi fare	the rate never changes	night tariff, waiting time
ball thrown up	no air resistance	light or fast objects
revenue = price × sales	sales fall linearly with price	very high or very low prices

A model is not right or wrong. It is useful over some range of inputs, and the honest answer says which range.

02 Worked examples

EXAMPLE 1 A rectangle has perimeter 40 cm. Find the greatest possible area.

① Let the width be x cm.

Length $20 - x$.

② $A(x) = x(20 - x) = 20x - x^2$

Domain $0 < x < 20$.

③ $= -(x - 10)^2 + 100$

Vertex at $x = 10$.

④ A square, side 10 cm.

ANSWER

100 cm², when the rectangle is a 10 cm square

EXAMPLE 2 A shop sells 200 notebooks a week at 12 000 so'm. Each 1 000 so'm rise loses 10 sales. What price maximises revenue?

① Let there be x rises of 1 000.

Price $12000 + 1000x$, sales $200 - 10x$.

② $R(x) = (12000 + 1000x)(200 - 10x)$

③ $= 2400000 + 80000x - 10000x^2$

④ $= -10000(x - 4)^2 + 2560000$

$x = 4$.

ANSWER

Price 16 000 so'm; revenue 2 560 000 so'm

EXAMPLE
3**A ball is thrown up: $h(t) = 20t - 5t^2$ metres after t seconds. Find the greatest height and when it lands.**

① $h(t) = -5(t - 2)^2 + 20$
Complete the square.

② Greatest height 20 m at $t = 2\text{ s}$.

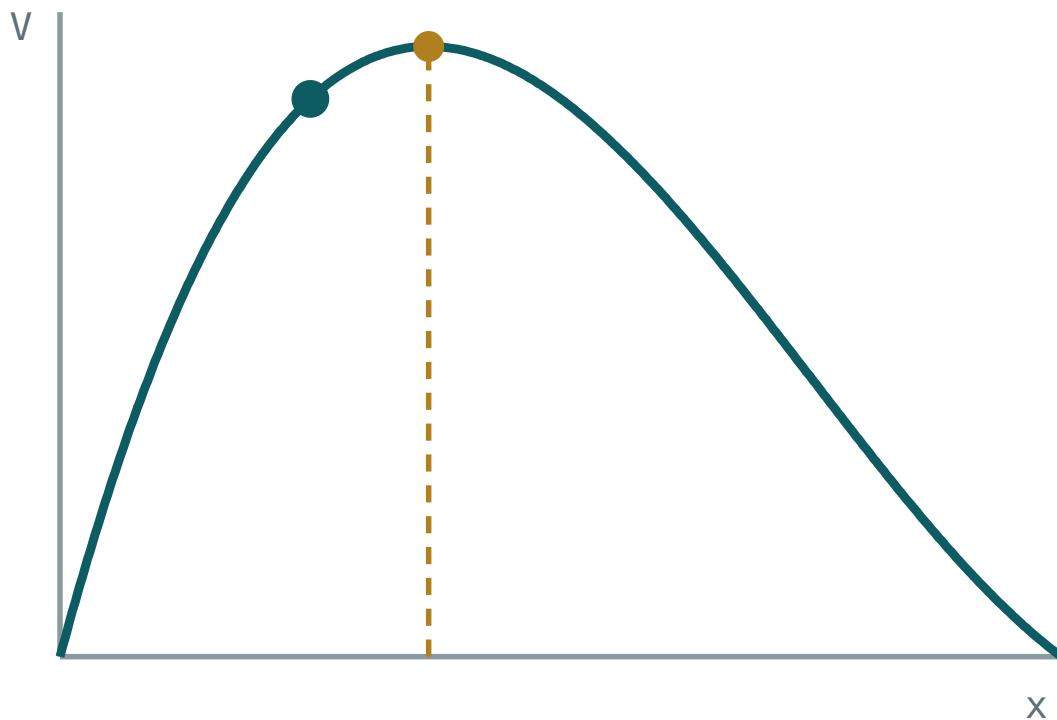
③ $20t - 5t^2 = 0 \Rightarrow 5t(4 - t) = 0$

④ $t = 0$ or $t = 4$.
Domain $0 \leq t \leq 4$.

ANSWER 20 m at $t = 2$; lands at $t = 4\text{ s}$ **03 Interactive model**

Change the total length of fence and watch the optimum width move with it.

The largest box



x 2

 $V(x)$ 384

greatest V 420.1

at x = 2.94

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Mathematical model	Matematik model	Математическая модель
Variable	O'zgaruvchi	Переменная
Constraint	Cheklov	Ограничение
Constant term	O'zgarmas had	Свободный член
Rate of change	O'zgarish tezligi	Скорость изменения
Maximum value	Eng katta qiymat	Наибольшее значение
Optimal value	Optimal qiymat	Оптимальное значение
Interpretation	Talqin	Интерпретация
Assumption	Faraz	Допущение
Limitation of a model	Model cheklovi	Ограниченность модели

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $C = 8000 + 2500d$, the number 2500 is:

the fixed charge

the rate per km

the total

the domain

For a negative quadratic, the vertex gives:

the minimum

the maximum

a root

the intercept

A rectangle of fixed perimeter has greatest area when it is:

long and thin

a square

a golden rectangle

it has no maximum

The domain of a length variable is normally:

$$\mathbb{R}$$

$$x > 0$$

$$x \neq 0$$

$$x \leq 0$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A taxi charges 9 000 plus 3 000 a km. Write $C(d)$

ANSWER $C = 9000 + 3000d$

2. Cost of 7 km on that model

ANSWER 30 000 so'm

3. A rectangle is x by $10 - x$. Write the area

ANSWER $A = 10x - x^2$

4. Domain of that area function

ANSWER $0 < x < 10$

5. $h(t) = 10t - 5t^2$; when is $h = 0$?

ANSWER $t = 0, 2$

6. Greatest value of $A = -(x - 6)^2 + 36$

ANSWER 36

7. Where does it occur?

ANSWER $x = 6$

Medium · the standard question

7 PROBLEMS

1. Perimeter 24 cm; maximise the area

ANSWER 36 cm^2 , a 6 cm square

2. A card holds 25 000 and loses 400 a minute. When is it empty?

ANSWER 62.5 minutes

3. $h(t) = 30t - 5t^2$; greatest height

ANSWER 45 m at $t = 3$

4. Two numbers add to 20. Maximise their product

ANSWER 100, at 10 and 10

5. Fence 100 m, pen against a wall. Maximise the area

ANSWER 1250 m^2 with $x = 25$

6. A firm sells $100 - 2p$ items at price p . Write the revenue

ANSWER $R = 100p - 2p^2$

7. Maximise it

ANSWER $p = 25$, $R = 1250$

Hard · stretch and reason

7 PROBLEMS

1. Fence 80 m against a wall, but a 4 m gate is free. Maximise the area

ANSWER $x = 21$, area 882 m^2

2. Cost $C = x^2 - 40x + 500$. Find the least cost and the x

ANSWER 100 at $x = 20$

3. $h(t) = 1.5 + 12t - 5t^2$. When does the ball reach 8 m?

ANSWER $t \approx 0.71$ and $t \approx 1.69 \text{ s}$

4. Two numbers differ by 6. Minimise the sum of their squares

ANSWER 18, at 3 and -3

5. A 20 cm wire is cut in two and each part bent into a square. Minimise the total area

ANSWER cut in half; 12.5 cm^2

6. Sales fall by 5 for every 500 so'm rise from 10 000 where 300 sell. Maximise revenue

ANSWER price 20 000, revenue 3 000 000

7. Name one assumption in the previous model and where it fails

ANSWER Sales fall linearly — false at very high prices, where they hit zero

07 Homework

Homework — 5 tasks

Every answer needs units and a sentence, not just a number.

1. A plumber charges 50 000 so'm plus 30 000 an hour. Write the model, find the cost of a $3\frac{1}{2}$ hour job, and state the domain.
2. A rectangular garden has perimeter 36 m. Find the dimensions giving the greatest area.
3. $h(t) = 25t - 5t^2$. Find the greatest height, the time it occurs and when the ball lands.
4. A stall sells 80 samsa a day at 8 000 so'm. Each 500 so'm rise loses 4 sales. Find the revenue-maximising price.

5. Name one assumption in task 4 and say where it would break down.

Control work 2, and the quarter review

The whole of Chapter 1 in one paper, then an hour drawing the quarter together into a single map.

LESSON 26–27

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 2

P1 · CHAPTERS 2–3 REVIEW

LESSON CLOCK

3

42'

10'

20'

10'

5'

Instructions	3 min
The paper	42 min
Answers on the board	10 min
Concept map in pairs	20 min
Targets for Quarter II	10 min
Homework	5 min

LEARNING OBJECTIVES

- Apply every idea of the quarter in one assessment.
- Choose the right tool without being told which section a question comes from.
- Build a concept map linking function, domain, range, property and transformation.
- Set a personal target for Quarter II.

TEXTBOOK

National	pp. 89–92
Cambridge	Pure Mathematics 1, pp. 56, 80
Workbook	Control paper B

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS
1	Domain and range of $f(x) = \frac{3}{x-2}$	3
2	$f = 2x - 1, g = x^2$: find $fg(x), gf(x)$ and solve $fg(x) = gf(x)$	5
3	Find f^{-1} for $f(x) = \frac{x+1}{x-2}$ and state its domain	4
4	Test $x^4 - x^2$ and $x^3 + x$ for parity, showing $f(-x)$	4
5	Sign intervals of $f(x) = x^3 - 9x$	4
6	Describe the transformation from $y = x^2$ to $y = 2(x+1)^2 - 5$	4
7	A 48 m fence against a wall: find the greatest area, with the domain stated	6

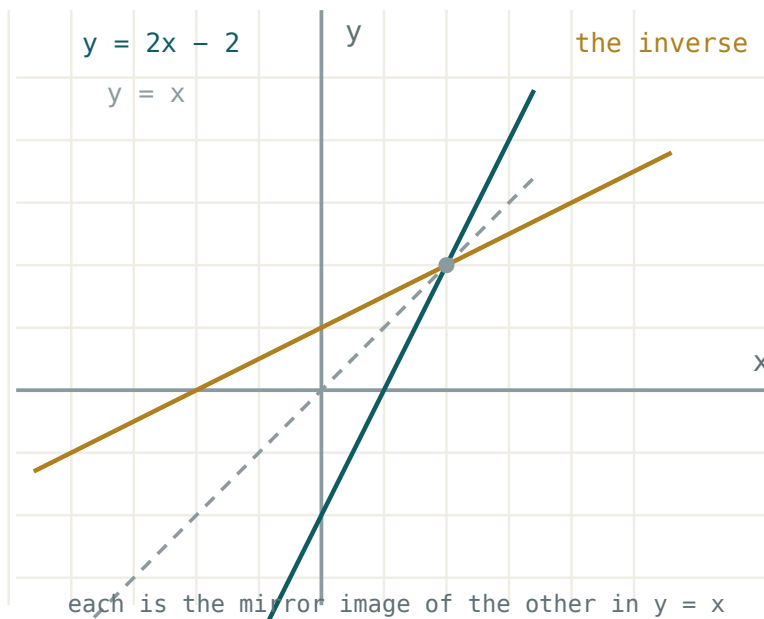
COMMAND WORDS

State — no working needed. **Show that** — every line must be there. **Find** — working plus answer.
Hence — you must use the previous part.

The concept map

In pairs, one A3 sheet. Six boxes, and every link labelled with a sentence:

- **mapping** → **function** — “exactly one output per input”
- **function** → **domain and range** — “what may go in, what comes out”
- **domain** → **one-to-one** — “restrict it and the function can be inverted”
- **one-to-one** → **inverse** — “reflect in $y = x$ ”
- **function** → **properties** — “zeros, sign, monotonic, parity, bounds”
- **function** → **transformations** — “inside the bracket works backwards”



The single picture that ties four boxes of the map together.

Two links must be the pair's own, not from this list.

Looking forward

Quarter II opens with rational and irrational equations. Everything in it depends on one habit built this quarter: **state the domain before you solve**. An irrational equation manufactures false roots precisely when the domain is ignored.

WRITE ONE TARGET

One sentence, in the exercise book, dated. Not “work harder” — something checkable, such as “state the domain in every question, in writing, before I start solving”.

02 Worked examples

EXAMPLE 1 Model answer, Q3: find f^{-1} for $f(x) = \frac{x+1}{x-2}$.

① $y(x-2) = x+1$

Clear the fraction.

② $yx - 2y = x + 1$

③ $x(y-1) = 2y+1$

Collect the x terms.

④ $x = \frac{2y+1}{y-1}$

Swap the letters.

ANSWER

$$f^{-1}(x) = \frac{2x+1}{x-1}, \text{ domain } x \neq 1$$

EXAMPLE 2 Model answer, Q7: 48 m of fence, three sides of a rectangle against a wall.

- ① Let each of the two equal sides be x m.

Third side $48 - 2x$.

- ② $A(x) = x(48 - 2x) = 48x - 2x^2$

Domain $0 < x < 24$.

- ③ $= -2(x - 12)^2 + 288$

- ④ $x = 12$, other side 24.

ANSWER

288 m^2 , a 12 m by 24 m pen

EXAMPLE 3 Model answer, Q5: sign intervals of $f(x) = x^3 - 9x$.

- ① $x(x - 3)(x + 3)$

- ② Zeros $-3, 0, 3$.

Four intervals.

- ③ Test $-4, -1, 1, 4$: $-28, 8, -8, 28$.

ANSWER

$f > 0$ on $(-3, 0) \cup (3, \infty)$; $f < 0$ on $(-\infty, -3) \cup (0, 3)$

03 Interactive model

Run through the six boxes of the map with the class before the pairs start.

The quarter in ten questionsDomain of $\sqrt{3x - 2}$:

$x \neq 2$

$x > 2$

$x \neq 3$

$\mathbb{R}$

Range of $\sqrt{3x - 2}$:

$y \neq 0$

$y \neq 2$

$y \neq 3$

$\mathbb{R}$

 $f = 2x - 1, g = x^2; fg(x)$ is:

$2x^2 - 1$

$(2x - 1)^2$

$4x^2 - 1$

$2x^2 + 1$

$x^4 - x^2$ is:

even

odd

neither

both

$x^3 + x$ is:

even

odd

neither

both

$y = 2(x + 1)^2 - 5$ is $y = x^2$:

left 1, stretch 2, down 5

right 1, stretch 2, down 5

left 1, stretch 2, up 5

left 2, stretch 1, down 5

The greatest area from 48 m against a wall is:

144 m^2

288 m^2

576 m^2

192 m^2

A function has an inverse exactly when it is:

The zeros of $x^3 - 9x$ are:

$f(x) = x^2$ is decreasing on:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Revision	Takrorlash	Повторение
Concept map	Tushunchalar xaritasi	Карта понятий
Summative assessment	Yakuniy baholash	Итоговое оценивание
Criterion	Mezon	Критерий
Target	Maqsad	Цель
Self-assessment	O'z-o'zini baholash	Самооценка
Command word	Topshiriq so'zi	Командное слово
Quarter	Chorak	Четверть

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Show that” means:

answer only

every line of working

a sketch

use a calculator

Before solving an equation with a fraction you should:

multiply out

state the domain

square both sides

guess

The vertex of $y = 2(x + 1)^2 - 5$ is:

$(1, -5)$

$(-1, -5)$

$(-1, 5)$

$(2, -5)$

A useful personal target is:

“work harder”

“be better at algebra”

“state the domain in writing every time”

“get 100%”

06

Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $\frac{3}{x-2}$

ANSWER $x \neq 2$

2. Zeros of $x^3 - 9x$

ANSWER $0, \pm 3$

3. Is x^4 even or odd?

ANSWER even

4. Inverse of $f(x) = x + 9$

ANSWER $x - 9$

5. Vertex of $y = (x + 1)^2 - 5$

ANSWER $(-1, -5)$

6. $f = 2x, g = x + 1; fg(2)$

ANSWER 6

7. Range of $y = x^2 + 2$

ANSWER $y \geq 2$

Medium · the standard question

7 PROBLEMS

1. $f = 2x - 1, g = x^2$; solve $fg(x) = gf(x)$

ANSWER $x = 0$ or $x = 1$

2. Inverse of $\frac{x+1}{x-2}$

ANSWER $\frac{2x+1}{x-1}$

3. Sign intervals of $x^3 - 9x$

ANSWER > 0 on $(-3, 0) \cup (3, \infty)$

4. Describe $y = 2(x+1)^2 - 5$ from $y = x^2$

ANSWER left 1, stretch 2, down 5

5. 48 m against a wall — greatest area

ANSWER 288 m^2

6. Range of $\frac{3}{x-2}$

ANSWER $y \neq 0$

7. Parity of $x^3 + x$

ANSWER odd

Hard · stretch and reason

7 PROBLEMS

1. Domain of $\frac{\sqrt{x+2}}{x^2-9}$

ANSWER $x \geq -2, x \neq 3$

2. $f = \sqrt{x}, g = x - 4$; find fg and gf with domains

ANSWER $\sqrt{x-4}$ on $x \geq 4$; $\sqrt{x} - 4$ on $x \geq 0$

3. Restrict $f(x) = x^2 + 6x$ to invert it, then find f^{-1}

ANSWER $x \geq -3$; $f^{-1}(x) = -3 + \sqrt{x+9}$

4. Show that $f(x) = \frac{x}{x^2+1}$ is odd and bounded

ANSWER $f(-x) = -f(x)$; $|f| \leq 0.5$

5. 60 m of fence, two equal pens side by side against a wall. Maximise the total area

ANSWER $x = 10$, area 300 m^2

6. Find a, b with $y = a(x-2)^2 + b$ through $(0, 7)$ and $(2, -1)$

ANSWER $a = 2, b = -1$

7. Explain why f^{-1} has the domain of the range of f

ANSWER Inputs and outputs exchange roles

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter II.

1. Rewrite every control-work question that lost a mark, in full.
2. Finish the concept map, including the two links your pair invented.
3. Write your target for Quarter II in one checkable sentence, and date it.

4. Domain and range of $f(x) = \sqrt{9 - x^2}$, with a sketch.